Advanced Linear Algebra I

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Introduction

Linear algebra is often first encountered as a collection of computational techniques: solving systems of linear equations, performing matrix operations, evaluating determinants, and finding eigenvalues. These computations are indispensable, but they represent only one layer of the subject. Behind them lies a structural theory of vector spaces, linear maps, changes of coordinates, invariants, and canonical representations.
The purpose of these notes is to organize this theory into a sequence that begins with concrete computation and gradually moves toward structural and abstract viewpoints. Because the material naturally extends far beyond what can reasonably be treated as a single course, the notes are divided into three volumes:

$$\boxed{\text{Advanced Linear Algebra I} \longrightarrow \text{Advanced Linear Algebra II} \longrightarrow \text{Algebraic Frontiers.}}$$

These three volumes are not intended to represent three equally abstract levels of the same subject. They have different roles.

0.1 The Three-Volume Structure

Advanced Linear Algebra I develops the foundations of finite-dimensional linear algebra. It begins with sets, maps, and the basic algebraic language needed throughout the subject, and then moves through systems of linear equations, matrices, vector spaces, and linear maps. Determinants introduce an important invariant of linear operators, while characteristic polynomials and eigenvalues lead to the study of invariant directions. The volume ends with diagonalization and basic invariant-subspace decompositions.
Its progression may be summarized as

$$\boxed{\text{linear equations} \longrightarrow \text{matrices} \longrightarrow \text{vector spaces} \longrightarrow \text{linear maps}}$$

followed by

$$\boxed{\text{determinants} \longrightarrow \text{eigenvalues} \longrightarrow \text{diagonalization.}}$$

The central transition is from computation to structure. A matrix is not regarded merely as a rectangular array of numbers, but as the coordinate representation of a linear map. Once this distinction is made, change of basis becomes similarity, and the natural objective is to identify properties that remain unchanged when coordinates are changed.
Advanced Linear Algebra II continues from the point where ordinary diagonalization becomes insufficient. The emphasis remains on finite-dimensional linear algebra at the undergraduate level, but the objects and decompositions become more refined. Jordan normal form describes operators that do not possess an eigenbasis. Inner product spaces introduce orthogonality and the Gram–Schmidt process, leading to orthogonal and unitary transformations and to the spectral theory of symmetric and Hermitian matrices. Bilinear forms, quadratic forms, congruence, and related canonical forms provide another classification problem, while matrix factorizations such as the singular value decomposition reveal geometric and computational structure that similarity alone cannot capture.
Thus the second volume studies several different ways of simplifying a linear object:

$$\boxed{\text{similarity,} \quad \text{orthogonal/unitary equivalence,} \quad \text{congruence,} \quad \text{matrix factorization.}}$$

The guiding question changes from "Can this operator be diagonalized?" to the broader question "What is the simplest form of this linear object under the appropriate notion of equivalence?"

Algebraic Frontiers has a different purpose. Some topics that arise naturally in an honors algebra course already belong to the language of later courses in abstract algebra, geometry, and algebraic structures. Modules and quotient modules generalize vector spaces beyond fields; exact sequences organize relations among algebraic objects; tensor products provide a universal language for multilinear constructions. Further topics connect linear algebra with symplectic and Hamiltonian structures, Lie algebras, and commutative algebra.
These subjects are therefore separated from the two volumes of Advanced Linear Algebra. The separation is intentional:

Advanced Linear Algebra I–II finite-dimensional undergraduate linear algebra

while

Algebraic Frontiers extensions toward more advanced algebraic theories.

The third volume is consequently best regarded as a record and reference for material beyond the core undergraduate sequence, rather than as a prerequisite for understanding the first two volumes.

0.2 The Logic of This Volume

This first volume begins with the language of sets and maps because linear algebra is fundamentally a theory of structures and the maps between them. Linear systems then provide the concrete computational origin of the subject, and matrices encode these systems in a form suitable for systematic manipulation.
Vector spaces abstract the common linear structure behind coordinate spaces, polynomial spaces, matrix spaces, and solution spaces. Linear maps then become the natural transformations between these spaces. After bases are chosen, every finite-dimensional linear map is represented by a matrix:

$$[T(v)]_{\mathcal{C}} = [T]_{\mathcal{C}\leftarrow\mathcal{B}}[v]_{\mathcal{B}}.$$

This formula provides one of the central organizing principles of the volume. The vector v and the linear map T are intrinsic objects, while their coordinate vector and matrix representation depend on the chosen bases.
For a linear operator $T : V \to V$, changing the basis replaces its matrix A by

$$P^{-1}AP.$$

The resulting notion of similarity leads naturally to the search for similarity invariants. Rank, determinant, trace, characteristic polynomial, and eigenvalues provide increasingly refined information about the underlying operator.
Eigenvectors identify one-dimensional invariant subspaces:

$$T(v) = \lambda v.$$

If enough such directions exist to form a basis, the operator becomes diagonal:

$$[T]_{\mathcal{B}} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Diagonalization therefore provides the natural endpoint of the first volume. It represents the first major classification result of linear algebra: under favorable conditions, a complicated matrix can be replaced, after a change of coordinates, by independent scalar actions along suitable directions.

It also exposes the problem that motivates the second volume. Not every operator has enough eigenvectors to be diagonalizable. The failure of diagonalization is not the end of the theory; rather, it indicates that more refined canonical forms and additional geometric structures are needed.

Accordingly, this volume establishes the language and structural foundation on which the second volume will build:

$$\boxed{\text{representation} \longrightarrow \text{invariance} \longrightarrow \text{simplification} \longrightarrow \text{classification.}}$$

Chapter 1

Sets, Maps, and Basic Algebraic Structures

1.1 Sets and Basic Notation

Linear algebra is built on a small collection of basic mathematical objects and notations. We first fix the language that will be used throughout the rest of the notes.

A *set* is a collection of objects. If x is an element of a set X, we write $x \in X$; otherwise we write $x \notin X$.

If every element of A is also an element of B, then A is a subset of B, written $A \subseteq B$. If $A \subseteq B$ and $A \neq B$, then A is a proper subset of B.

The empty set is denoted by $\varnothing$.

Frequently used number systems include

$$\mathbb{N} = \{1, 2, 3, \ldots\}, \qquad \mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\},$$

together with the rational numbers $\mathbb{Q}$, the real numbers $\mathbb{R}$, and the complex numbers $\mathbb{C}$.

For a set X, its cardinality is denoted by $|X|$ or $\mathrm{Card}(X)$ when X is finite.

For two sets A and B, their union and intersection are

$$A \cup B = \{x : x \in A \text{ or } x \in B\},$$

and

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

The set difference is

$$A \setminus B = \{x \in A : x \notin B\}.$$

Definition 1.1.1. The power set of X, denoted by $\mathcal{P}(X)$, is the set of all subsets of X:

$$\mathcal{P}(X) = \{A : A \subseteq X\}.$$

If X has n elements, then

$$|\mathcal{P}(X)| = 2^n.$$

1.2 Cartesian Products

Definition 1.2.1. For two sets X and Y, their Cartesian product is

$$X \times Y = \{(x, y) : x \in X,\ y \in Y\}.$$

More generally,

$$X_1 \times \cdots \times X_n = \{(x_1, \ldots, x_n) : x_i \in X_i\}.$$

If $X_1 = \cdots = X_n = X$, we write

$$X^n = \underbrace{X \times \cdots \times X}_{n \text{ times}}.$$

In particular,

$$\mathbb{R}^n = \{(x_1, \ldots, x_n) : x_i \in \mathbb{R}\},$$

and similarly for $\mathbb{Q}^n$ and $\mathbb{C}^n$.

Remark 1.2.2. Cartesian products provide the natural set-theoretic model for coordinate vectors. Later, K^n will become the fundamental example of an n-dimensional vector space over a field K.

1.3 Maps

Definition 1.3.1. A map

$$f : X \to Y$$

assigns to every $x \in X$ a unique element $f(x) \in Y$.
The set X is called the *domain* of f, and Y is called the *codomain*.

For $x \in X$, the element $f(x)$ is called the image of x under f.
If $A \subseteq X$, the image of A under f is

$$f(A) = \{f(a) : a \in A\}.$$

In particular, the image of the entire map is

$$\mathrm{Im}(f) = f(X) = \{f(x) : x \in X\}.$$

If $B \subseteq Y$, the inverse image or preimage of B is

$$f^{-1}(B) = \{x \in X : f(x) \in B\}.$$

Remark 1.3.2. The notation $f^{-1}(B)$ for a preimage is defined for every map f. It does not require f to possess an inverse map.

For a single element $y \in Y$,

$$f^{-1}(\{y\}) = \{x \in X : f(x) = y\}.$$

1.4 Injective, Surjective, and Bijective Maps

Definition 1.4.1. A map $f : X \to Y$ is *injective* if

$$f(x_1) = f(x_2)$$

implies $x_1 = x_2$.
Equivalently, distinct elements of X have distinct images.

Definition 1.4.2. A map $f : X \to Y$ is *surjective* if for every $y \in Y$, there exists $x \in X$ such that

$$f(x) = y.$$

Equivalently,

$$\mathrm{Im}(f) = Y.$$

Definition 1.4.3. A map is *bijective* if it is both injective and surjective.

Example 1.4.4. Consider

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2.$$

This map is neither injective nor surjective: $f(1) = f(-1)$, and negative real numbers are not in its image.
If instead

$$g : [0, \infty) \to [0, \infty), \qquad g(x) = x^2,$$

then g is bijective.

Proposition 1.4.5. A map $f : X \to Y$ has an inverse map

$$f^{-1} : Y \to X$$

if and only if f is bijective.

If f is bijective, then

$$f^{-1}(f(x)) = x$$

for every $x \in X$, and

$$f(f^{-1}(y)) = y$$

for every $y \in Y$.

1.5 Composition of Maps

Let

$$f : X \to Y, \qquad g : Y \to Z.$$

Definition 1.5.1. The composition of f and g is the map

$$g \circ f : X \to Z$$

defined by

$$(g \circ f)(x) = g(f(x)).$$

Composition is associative:

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

In general,

$$g \circ f \neq f \circ g.$$

Definition 1.5.2. The identity map on a set X is

$$\mathrm{id}_X : X \to X, \qquad \mathrm{id}_X(x) = x.$$

For every map $f : X \to Y$,

$$f \circ \mathrm{id}_X = f, \qquad \mathrm{id}_Y \circ f = f.$$

If $f : X \to Y$ is bijective, then its inverse satisfies

$$f^{-1} \circ f = \mathrm{id}_X, \qquad f \circ f^{-1} = \mathrm{id}_Y .$$

1.6 Equivalence Relations

Definition 1.6.1. A relation $\sim$ on a set X is called an equivalence relation if it satisfies:

1. reflexivity:

   $$x \sim x$$

   for every $x \in X$;

2. symmetry:

   $$x \sim y \Longrightarrow y \sim x;$$

3. transitivity:

   $$x \sim y, \quad y \sim z \Longrightarrow x \sim z.$$

For $x \in X$, its equivalence class is

$$[x] = \{y \in X : y \sim x\}.$$

Proposition 1.6.2. Two equivalence classes are either identical or disjoint.

Thus an equivalence relation partitions X into disjoint equivalence classes.
The set of all equivalence classes is denoted by

$$X/\sim.$$

Remark 1.6.3. Equivalence relations will later appear naturally when different matrices are regarded as representing the same underlying linear structure, for example under matrix equivalence or similarity.

1.7 Binary Operations

Definition 1.7.1. A binary operation on a set X is a map

$$* : X \times X \to X.$$

For $x, y \in X$, the image of (x, y) is written $x * y$.

Examples include addition on $\mathbb{Z}$, multiplication on $\mathbb{R}$, and matrix addition on sets of matrices of fixed size.
A binary operation is associative if

$$(x * y) * z = x * (y * z)$$

for all $x, y, z \in X$, and commutative if

$$x * y = y * x$$

for all $x, y \in X$.
An element $e \in X$ is an identity element if

$$e * x = x * e = x$$

for all $x \in X$.
If an identity element exists, an element $x \in X$ is invertible if there exists $y \in X$ such that

$$x * y = y * x = e.$$

1.8 Fields

The scalar systems used in linear algebra must support addition, subtraction, multiplication, and division by nonzero elements. These properties are formalized by the notion of a field.

Definition 1.8.1. A field K is a set equipped with addition and multiplication such that:

1. K is an abelian group under addition;
2. $K \setminus \{0\}$ is an abelian group under multiplication;

3. multiplication distributes over addition:

$$a(b+c) = ab + ac$$

for all $a, b, c \in K$.

The standard examples are

$$\mathbb{Q}, \qquad \mathbb{R}, \qquad \mathbb{C}.$$

For $a \in K$, its additive inverse is denoted by $-a$. For $a \neq 0$, its multiplicative inverse is denoted by a^{-1}.
Thus subtraction and division are defined by

$$a - b = a + (-b)$$

and

$$\frac{a}{b} = ab^{-1}, \qquad b \neq 0.$$

Proposition 1.8.2. Let K be a field. Then for $a, b \in K$,

$$ab = 0$$

implies

$$a = 0 \qquad \text{or} \qquad b = 0.$$

Proof. Suppose $a \neq 0$. Since K is a field, a^{-1} exists. From $ab = 0$,

$$a^{-1}(ab) = a^{-1}0.$$

Hence $b = 0$. □

1.9 Complex Numbers

Because complex scalars will appear naturally in eigenvalue problems, we recall the basic structure of $\mathbb{C}$.
Every complex number can be written uniquely as

$$z = a + bi, \qquad a, b \in \mathbb{R},$$

where $i^2 = -1$.
The real and imaginary parts are

$$\mathrm{Re}(z) = a, \qquad \mathrm{Im}(z) = b.$$

The complex conjugate of $z = a + bi$ is

$$\overline{z} = a - bi.$$

It satisfies

$$\overline{z + w} = \overline{z} + \overline{w},$$

$$\overline{zw} = \overline{z}\,\overline{w},$$

and

$$\overline{\overline{z}} = z.$$

The modulus of z is

$$|z| = \sqrt{z\overline{z}} = \sqrt{a^2 + b^2}.$$

For $z \neq 0$,

$$z^{-1} = \frac{\overline{z}}{|z|^2}.$$

Remark 1.9.1. Complex conjugation plays no essential role in the elementary theory of linear equations, but it becomes fundamental when complex inner product spaces and unitary transformations are introduced later.

1.10 The Algebraic Setting of Linear Algebra

The constructions introduced in this chapter provide the language used throughout the rest of linear algebra.

A typical linear algebra problem consists of a set V whose elements can be added and multiplied by scalars from a field K. Maps between such sets will later be required to preserve these operations.

The basic pattern is therefore

$$\text{sets} \longrightarrow \text{maps} \longrightarrow \text{operations} \longrightarrow \text{fields} \longrightarrow \text{vector spaces}.$$

The next chapter begins the concrete development of this theory with systems of linear equations and matrices.

Chapter 2

Systems of Linear Equations

2.1 Linear Equations and Linear Systems

Let K be a field. A linear equation in the unknowns $x_1, \ldots, x_n$ over K has the form

$$a_1x_1 + \cdots + a_nx_n = b,$$

where $a_1, \ldots, a_n, b \in K$.
A system of m linear equations in n unknowns is

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2, \\ \qquad\qquad\qquad\qquad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m. \end{cases}$$

A solution of the system is an ordered n-tuple $(x_1, \ldots, x_n) \in K^n$ satisfying every equation simultaneously. The set of all solutions is called the solution set of the system.

Definition 2.1.1. A linear system is called *consistent* if it has at least one solution, and *inconsistent* if it has no solution.

A consistent system may have either a unique solution or infinitely many solutions.

Definition 2.1.2. A linear system is called *homogeneous* if all constant terms are zero:

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = 0, \\ \qquad \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = 0. \end{cases}$$

Otherwise it is called *nonhomogeneous*.

Every homogeneous system has the trivial solution

$$x_1 = \cdots = x_n = 0.$$

Thus a homogeneous system is always consistent. The important question is whether it possesses nontrivial solutions.

Example 2.1.3. Consider

$$\begin{cases} x + 2y - z = 1, \\ 2x + 4y - 2z = 2. \end{cases}$$

The second equation is twice the first, so it contributes no additional constraint. The system therefore has infinitely many solutions.

The first fundamental problem of linear algebra is to determine the solution set of such systems systematically.

2.2 Coefficient and Augmented Matrices

A linear system can be recorded more efficiently by listing only its coefficients.
For

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1, \\ \quad \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m, \end{cases}$$

the coefficient matrix is

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

The augmented matrix is

$$(A \,|\, b) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right).$$

The vertical line is only a visual separator between the coefficients of the unknowns and the constant terms.
For example,

$$\begin{cases} x + 2y - z = 3, \\ 2x - y + 3z = 1 \end{cases}$$

has augmented matrix

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & -1 & 3 & 1 \end{array}\right).$$

The essential observation is that solving the original system can be replaced by manipulating its augmented matrix.

2.3 Elementary Row Operations

Certain transformations of a system change its appearance without changing its solution set.
The three elementary row operations are:

1. interchange two rows:
$$R_i \leftrightarrow R_j;$$
2. multiply one row by a nonzero scalar c:
$$R_i \mapsto cR_i, \qquad c \neq 0;$$
3. add a scalar multiple of one row to another:
$$R_j \mapsto R_j + cR_i.$$

Theorem 2.3.1. Elementary row operations preserve the solution set of a linear system.

Proof. Interchanging two equations clearly does not change the conditions that a solution must satisfy.
Multiplying an equation by $c \neq 0$ gives an equivalent equation, since multiplication by c^{-1} recovers the original equation.
Finally, replacing one equation by itself plus c times another equation preserves every solution of the original system. Conversely, the original equation can be recovered by subtracting c times the unchanged equation. Hence the transformed system has exactly the same solutions. □

Definition 2.3.2. Two systems of linear equations are called equivalent if they have the same solution set.

Thus elementary row operations transform a linear system into an equivalent one.

Definition 2.3.3. Two matrices are called row equivalent if one can be obtained from the other by a finite sequence of elementary row operations.

For solving systems, the goal is therefore to replace a complicated augmented matrix by a simpler row-equivalent matrix.

2.4 Row Echelon Forms

Definition 2.4.1. A matrix is in *row echelon form* if:

1. all zero rows, if any, occur below all nonzero rows;
2. the first nonzero entry of each nonzero row lies strictly to the right of the first nonzero entry of the preceding row;
3. all entries below each leading entry are zero.

The first nonzero entry in a nonzero row is called a *pivot* or *leading entry.* Its position is called a pivot position.
For example,

$$\begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 4 & 2 \\ 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

is in row echelon form.

Definition 2.4.2. A matrix is in *reduced row echelon form* if:

1. it is in row echelon form;
2. every pivot is equal to 1;
3. every pivot is the only nonzero entry in its column.

For example,

$$\begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

is in reduced row echelon form.

Theorem 2.4.3. Every matrix is row equivalent to a matrix in row echelon form.

Theorem 2.4.4. Every matrix is row equivalent to a unique reduced row echelon form.

The uniqueness of the reduced row echelon form is important: although different sequences of elementary row operations may be used, the final reduced row echelon form is always the same.

2.5 Gaussian Elimination

The systematic procedure for transforming a matrix into row echelon form is called *Gaussian elimination*.
The basic strategy is:

1. locate the leftmost column containing a nonzero entry;
2. choose a nonzero entry in that column as a pivot;
3. interchange rows if necessary to move the pivot upward;
4. eliminate all entries below the pivot;
5. repeat the procedure on the remaining rows and columns.

Continuing the elimination upward and normalizing every pivot to 1 produces reduced row echelon form. This extended procedure is called *Gauss–Jordan elimination*.

Example 2.5.1. Consider

$$\begin{cases} x + y + z = 6, \\ 2x - y + z = 3, \\ x + 2y - z = 2. \end{cases}$$

The augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 2 \end{array}\right).$$

Apply

$$R_2 \mapsto R_2 - 2R_1, \qquad R_3 \mapsto R_3 - R_1,$$

to obtain

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 1 & -2 & -4 \end{array}\right).$$

Interchanging the last two rows gives

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & -3 & -1 & -9 \end{array}\right).$$

Then

$$R_3 \mapsto R_3 + 3R_2$$

gives

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & -7 & -21 \end{array}\right).$$

Hence $z = 3$, then $y = 2$, and finally $x = 1$.

Gaussian elimination is not merely a computational trick. It reveals which variables are constrained independently and which equations are redundant.

2.6 Pivot Variables and Free Variables

Suppose an augmented matrix has been transformed into row echelon form.
A variable corresponding to a pivot column of the coefficient matrix is called a *pivot variable*. A variable corresponding to a nonpivot column is called a *free variable*.
For example, consider

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 4 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

The pivot variables are x_1 and x_3, while x_2 and x_4 are free.
The corresponding equations are

$$x_1 + 2x_2 - x_4 = 3, \qquad x_3 + 4x_4 = 2.$$

Let

$$x_2 = s, \qquad x_4 = t.$$

Then

$$x_1 = 3 - 2s + t, \qquad x_3 = 2 - 4t.$$

Therefore the complete solution is

$$(x_1, x_2, x_3, x_4) = (3 - 2s + t,\ s,\ 2 - 4t,\ t), \qquad s, t \in K.$$

Equivalently,

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 2 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ -4 \\ 1 \end{pmatrix}.$$

This form anticipates the later interpretation of solution sets using vector spaces.

2.7 Existence and Number of Solutions

Row echelon form makes consistency easy to determine.
A row of the form

$$\left(\begin{array}{ccccc} 0 & 0 & \cdots & 0 & c \end{array}\right), \qquad c \neq 0,$$

represents the impossible equation

$$0 = c.$$

Hence the system is inconsistent.

Theorem 2.7.1. A system of linear equations is consistent if and only if its row echelon form contains no row corresponding to an equation

$$0 = c, \qquad c \neq 0.$$

For a consistent system, there are two possibilities:

1. every variable is a pivot variable, in which case the solution is unique;
2. at least one variable is free, in which case there are infinitely many solutions when the scalar field is infinite.

Thus over $\mathbb{R}$ or $\mathbb{C}$, a linear system has exactly one of three possible outcomes:

no solution,	one solution,	infinitely many solutions.

2.8 Homogeneous Systems

Consider the homogeneous system

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = 0, \\ \quad \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = 0. \end{cases}$$

Because the zero tuple is always a solution, the system is necessarily consistent.

Theorem 2.8.1. A homogeneous linear system has a nontrivial solution if at least one variable is free.

In particular, if a homogeneous system has more unknowns than pivot positions, then it has a nontrivial solution.

Corollary 2.8.2. A homogeneous system of m linear equations in n unknowns with $n > m$ has a nontrivial solution.

Proof. There can be at most m pivot positions because there are only m rows. Since $n > m$, at least one of the n variables must be free. Assigning a suitable nonzero value to a free variable produces a nontrivial solution. □

This simple observation will later become a consequence of the dimension theory of vector spaces.

2.9 Homogeneous and Nonhomogeneous Solution Sets

There is a fundamental relationship between a consistent nonhomogeneous system and its associated homogeneous system.
Suppose x_p is one particular solution of a nonhomogeneous system, and let x_h be a solution of the associated homogeneous system. Then

$$x_p + x_h$$

is again a solution of the nonhomogeneous system.
Indeed, in matrix notation this observation will later be written as

$$A(x_p + x_h) = Ax_p + Ax_h = b + 0 = b.$$

Conversely, if x_1 and x_2 are two solutions of the same nonhomogeneous system, then their difference $x_1 - x_2$ solves the associated homogeneous system.
Therefore every solution has the form

$$\boxed{\text{general solution} = \text{particular solution} + \text{homogeneous solution.}}$$

Example 2.9.1. Suppose a system has one particular solution

$$x_p = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix},$$

while the associated homogeneous system has solutions

$$x_h = t \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \qquad t \in \mathbb{R}.$$

Then every solution of the nonhomogeneous system is

$$x = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

Later, after vector spaces have been introduced, the homogeneous solution set will be recognized as a subspace, while the solution set of a consistent nonhomogeneous system will be a translation of that subspace.

2.10 Linear Systems as the Starting Point of Linear Algebra

The theory developed in this chapter begins with an elementary question: how can a collection of linear equations be solved?
Gaussian elimination gives the computational answer. At the same time, the elimination process reveals several structural phenomena:

- some equations may be redundant;
- some variables may be determined by others;
- free variables generate families of solutions;
- homogeneous systems possess an internal linear structure;
- nonhomogeneous solution sets are obtained by translating homogeneous solution sets.

These observations suggest that linear equations should not be studied only as collections of formulas. Their coefficients can be organized into matrices, and their solution sets can eventually be understood as geometric and algebraic objects.
The next chapter therefore develops matrices as mathematical objects in their own right. Matrix operations, multiplication, invertibility, elementary matrices, block matrices, and rank will provide a more efficient language for the theory initiated here.

Chapter 3

Matrices

3.1 Matrices and Basic Operations

Let K be a field.

Definition 3.1.1. An $m \times n$ matrix over K is a rectangular array

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \qquad a_{ij} \in K.$$

The set of all $m \times n$ matrices over K is denoted by $M_{m\times n}(K)$.

If $m = n$, the matrix is called a square matrix of order n, and we write $M_n(K)$.
Two matrices $A = (a_{ij})$ and $B = (b_{ij})$ are equal if they have the same size and

$$a_{ij} = b_{ij}$$

for every i, j.
For $A, B \in M_{m\times n}(K)$, their sum is defined entrywise by

$$A + B = (a_{ij} + b_{ij}).$$

For $c \in K$, scalar multiplication is defined by

$$cA = (ca_{ij}).$$

The zero matrix is denoted by 0. It satisfies

$$A + 0 = A.$$

The additive inverse of A is

$$-A = (-a_{ij}),$$

so subtraction is defined by $A - B = A + (-B)$.
Matrix addition satisfies

$$A + B = B + A,$$

$$(A + B) + C = A + (B + C),$$

and

$$c(A + B) = cA + cB.$$

These operations already resemble the operations on ordinary vectors. Their abstract interpretation will be developed later when vector spaces are introduced.

3.2 Matrix Multiplication

Matrix multiplication is defined so that matrices can encode systems of linear equations and compositions of linear transformations.
Let

$$A = (a_{ij}) \in M_{m\times n}(K), \qquad B = (b_{ij}) \in M_{n\times p}(K).$$

Definition 3.2.1. The product $AB \in M_{m\times p}(K)$ is defined by

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

Thus the (i,j)-entry of AB is obtained by multiplying the i-th row of A with the j-th column of B.
For example,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax+by \\ cx+dy \end{pmatrix}.$$

This explains the matrix notation for a linear system:

$$Ax = b.$$

If

$$A = \begin{pmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{pmatrix}, \qquad x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

then

$$Ax = x_1 a_1 + \cdots + x_n a_n.$$

Thus matrix-vector multiplication forms a linear combination of the columns of A.

Theorem 3.2.2. Whenever the products are defined,

$$A(BC) = (AB)C.$$

Proof. The (i,j)-entry of $A(BC)$ is

$$\sum_k a_{ik}(BC)_{kj} = \sum_k a_{ik} \sum_l b_{kl} c_{lj} = \sum_{k,l} a_{ik} b_{kl} c_{lj}.$$

The (i,j)-entry of $(AB)C$ is

$$\sum_l (AB)_{il} c_{lj} = \sum_l \sum_k a_{ik} b_{kl} c_{lj} = \sum_{k,l} a_{ik} b_{kl} c_{lj}.$$

Hence $A(BC) = (AB)C$. □

Matrix multiplication is distributive:

$$A(B+C) = AB + AC$$

and

$$(A+B)C = AC + BC.$$

For $c \in K$,

$$c(AB) = (cA)B = A(cB).$$

However, matrix multiplication is generally not commutative:

$$AB \neq BA.$$

Example 3.2.3. Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Therefore $AB \neq BA$.

3.3 Identity Matrices and Matrix Powers

Definition 3.3.1. The identity matrix of order n is

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

For every $A \in M_{m\times n}(K)$,

$$I_m A = A = A I_n.$$

For a square matrix A, define

$$A^0 = I_n$$

and, for $k \geq 1$,

$$A^k = \underbrace{A \cdots A}_{k \text{ times}}.$$

Since matrix multiplication is associative,

$$A^r A^s = A^{r+s}$$

for nonnegative integers r, s.
If

$$p(x) = a_0 + a_1 x + \cdots + a_m x^m$$

is a polynomial, we may define

$$p(A) = a_0 I_n + a_1 A + \cdots + a_m A^m.$$

This construction will later become fundamental in the study of characteristic polynomials and linear operators.

3.4 Transpose and Special Matrices

Definition 3.4.1. For $A = (a_{ij}) \in M_{m\times n}(K)$, the transpose of A is the matrix $A^T \in M_{n\times m}(K)$ defined by

$$(A^T)_{ij} = a_{ji}.$$

The transpose satisfies

$$(A^T)^T = A,$$
$$(A + B)^T = A^T + B^T,$$
$$(cA)^T = cA^T,$$

and

$$(AB)^T = B^T A^T.$$

Definition 3.4.2. A square matrix A is symmetric if

$$A^T = A,$$

and skew-symmetric if

$$A^T = -A.$$

For every square matrix A over a field of characteristic not equal to 2,

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$

The first matrix is symmetric and the second is skew-symmetric.

Definition 3.4.3. A diagonal matrix has the form

$$D = \operatorname{diag}(d_1, \ldots, d_n) = \begin{pmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{pmatrix}.$$

A matrix is upper triangular if all entries below the main diagonal are zero, and lower triangular if all entries above the main diagonal are zero.
Triangular and diagonal matrices will later play a central role in simplifying linear operators.

3.5 Invertible Matrices

Definition 3.5.1. A square matrix $A \in M_n(K)$ is invertible if there exists $B \in M_n(K)$ such that

$$AB = BA = I_n.$$

The matrix B is called the inverse of A and is denoted by A^{-1}.

Proposition 3.5.2. If an inverse of A exists, then it is unique.

Proof. Suppose B and C are both inverses of A. Then

$$B = BI_n = B(AC) = (BA)C = I_nC = C.$$

Hence the inverse is unique. □

Theorem 3.5.3. If A and B are invertible matrices of the same order, then AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Also,

$$(A^{-1})^{-1} = A$$

and

$$(A^T)^{-1} = (A^{-1})^T.$$

For a positive integer k,

$$(A^k)^{-1} = (A^{-1})^k.$$

Theorem 3.5.4. For $A \in M_n(K)$, the following statements are equivalent:

1. A is invertible;
2. $Ax = 0$ has only the trivial solution;
3. $Ax = b$ has a unique solution for every $b \in K^n$;
4. the reduced row echelon form of A is I_n.

Later, these conditions will be supplemented by equivalent statements involving rank, linear independence, linear transformations, determinants, and eigenvalues.

3.6 Elementary Matrices

The elementary row operations introduced in Chapter 2 can themselves be represented by matrices.

Definition 3.6.1. An elementary matrix is a matrix obtained by performing one elementary row operation on an identity matrix.

If E is obtained from I_m by an elementary row operation, then for every $A \in M_{m\times n}(K)$, the product

$$EA$$

is obtained from A by performing the same row operation.
For example, if

$$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ c & 0 & 1 \end{pmatrix},$$

then left multiplication by E performs

$$R_3 \mapsto R_3 + cR_1.$$

Proposition 3.6.2. Every elementary matrix is invertible, and its inverse is also an elementary matrix.

Indeed, every elementary row operation can be reversed by another elementary row operation.

Theorem 3.6.3. A square matrix A is invertible if and only if it can be expressed as a product of elementary matrices.

Proof. Suppose A is invertible. Then row reduction transforms A into I_n, so there exist elementary matrices $E_1, \ldots, E_r$ such that

$$E_r \cdots E_1 A = I_n.$$

Hence

$$A = E_1^{-1} \cdots E_r^{-1},$$

which is a product of elementary matrices.
Conversely, every elementary matrix is invertible, and a product of invertible matrices is invertible. □

This theorem gives an algebraic interpretation of Gaussian elimination.

3.7 Computing the Inverse by Row Reduction

Suppose $A \in M_n(K)$. To determine whether A is invertible, consider the augmented matrix

$$(A \,|\, I_n).$$

If row reduction transforms A into I_n, the same sequence of row operations transforms I_n into A^{-1}:

$$(A \,|\, I_n) \longrightarrow (I_n \,|\, A^{-1}).$$

Indeed, if the row operations correspond to elementary matrices $E_1, \dots, E_r$, then

$$E_r \cdots E_1 A = I_n.$$

Therefore

$$E_r \cdots E_1 = A^{-1}.$$

Example 3.7.1. Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Begin with

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right).$$

Apply $R_2 \mapsto R_2 - 3R_1$:

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array}\right).$$

Then $R_2 \mapsto -\frac{1}{2}R_2$:

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right).$$

Finally, $R_1 \mapsto R_1 - 2R_2$:

$$\left(\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right).$$

Hence

$$A^{-1} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}.$$

3.8 Block Matrices

Large matrices can often be simplified by dividing them into smaller rectangular blocks. For example,

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},$$

where each A_{ij} is itself a matrix.
Block addition is performed componentwise whenever the block partitions are compatible:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} + \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} A+E & B+F \\ C+G & D+H \end{pmatrix}.$$

Block multiplication follows the same formal rule as ordinary matrix multiplication:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} AE+BG & AF+BH \\ CE+DG & CF+DH \end{pmatrix},$$

provided all products are defined.

Remark 3.8.1. Block multiplication is valid only when the partitions are compatible. The blocks themselves need not be square.

A particularly important example is a block diagonal matrix:

$$\begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_r \end{pmatrix}.$$

If every A_i is invertible, then

$$\begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_r \end{pmatrix}^{-1} = \begin{pmatrix} A_1^{-1} & & 0 \\ & \ddots & \\ 0 & & A_r^{-1} \end{pmatrix}.$$

Block matrices will later be useful for decompositions and canonical forms.

3.9 Matrix Equivalence

Row operations act on a matrix by multiplication from the left. Column operations can similarly be represented by multiplication from the right.

Definition 3.9.1. Two matrices $A, B \in M_{m\times n}(K)$ are equivalent if there exist invertible matrices $P \in M_m(K)$ and $Q \in M_n(K)$ such that

$$B = PAQ.$$

Equivalently, B can be obtained from A by a finite sequence of elementary row and column operations.
The purpose of matrix equivalence is to simplify a matrix while preserving its essential row-column structure.

Theorem 3.9.2. Every nonzero matrix $A \in M_{m\times n}(K)$ is equivalent to a matrix of the form

$$\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$$

for some integer r satisfying

$$1 \le r \le \min\{m, n\}.$$

The integer r is independent of the sequence of elementary operations used.

3.10 Rank of a Matrix

Definition 3.10.1. The rank of a matrix A, denoted by $\operatorname{rank}(A)$, is the number of nonzero rows in a row echelon form of A.

Equivalently, $\operatorname{rank}(A) = r$ if A is equivalent to

$$\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

Elementary row and column operations preserve rank.
For $A \in M_{m\times n}(K)$,

$$0 \le \operatorname{rank}(A) \le \min\{m, n\}.$$

If $A \in M_n(K)$, then

$$A \text{ is invertible } \iff \operatorname{rank}(A) = n.$$

Theorem 3.10.2. For every matrix A,

$$\operatorname{rank}(A) = \operatorname{rank}(A^T).$$

Theorem 3.10.3. For compatible matrices A and B,

$$\operatorname{rank}(AB) \leq \min\{\operatorname{rank}(A), \operatorname{rank}(B)\}.$$

Proposition 3.10.4. If P and Q are invertible, then

$$\operatorname{rank}(PAQ) = \operatorname{rank}(A).$$

Thus rank is an invariant of matrix equivalence.

3.11 Rank and Linear Systems

Rank gives a compact criterion for the solvability of a linear system.
Consider

$$Ax = b.$$

Let $(A \,|\, b)$ be the augmented matrix.

Theorem 3.11.1. The system $Ax = b$ is consistent if and only if

$$\operatorname{rank}(A) = \operatorname{rank}(A \,|\, b).$$

Proof. Row reduction preserves both the solution set and rank. The system is inconsistent precisely when its row echelon form contains a row

$$\begin{pmatrix} 0 & \cdots & 0 & 0 \,|\, c \end{pmatrix}, \qquad c \neq 0.$$

Such a row increases the rank of the augmented matrix without increasing the rank of the coefficient matrix. Therefore consistency is equivalent to

$$\operatorname{rank}(A) = \operatorname{rank}(A \,|\, b).$$

□

Suppose $A \in M_{m\times n}(K)$ and

$$\operatorname{rank}(A) = r.$$

If $Ax = b$ is consistent, then there are r pivot variables and $n - r$ free variables.
Therefore:

- if $r = n$, a consistent system has a unique solution;
- if $r < n$, a consistent system has free variables and hence infinitely many solutions over an infinite field.

For the homogeneous system $Ax = 0$, a nontrivial solution exists whenever

$$\operatorname{rank}(A) < n.$$

In particular, if $m < n$, then

$$\operatorname{rank}(A) \leq m < n,$$

so every homogeneous system of m equations in n unknowns has a nontrivial solution.

3.12 Row Rank and Column Rank

The rows and columns of a matrix can each be regarded as collections of coordinate vectors. Their precise interpretation as vectors in vector spaces will be developed in the next chapter.
For now, define the row rank of A as the maximum number of linearly independent rows, and the column rank as the maximum number of linearly independent columns.

Theorem 3.12.1. For every matrix A,

$$\text{row rank}(A) = \text{column rank}(A) = \text{rank}(A).$$

This theorem is one of the first indications that rank is not merely an artifact of Gaussian elimination. It measures an intrinsic property of the matrix.
Later, after vector spaces are introduced, the theorem will be written as

$$\text{rank}(A) = \dim R(A) = \dim C(A),$$

where $R(A)$ and $C(A)$ denote the row space and column space of A.

3.13 The Role of Matrices

Matrices first appeared in Chapter 2 as compact records of systems of linear equations. In this chapter they have become algebraic objects with their own operations and structural properties.
Several important ideas have emerged:

$$\text{Gaussian elimination} \longleftrightarrow \text{elementary matrices},$$

$$\text{unique solvability} \longleftrightarrow \text{invertibility},$$

and

$$\text{matrix simplification} \longleftrightarrow \text{rank}.$$

Yet matrices still depend on coordinates. The next step is therefore to identify the abstract structure underlying coordinate vectors and matrix equations.
This leads to the notion of a vector space. Linear combinations, linear independence, bases, and dimension will provide a coordinate-free language for many of the phenomena already observed in linear systems and matrices.

Chapter 4

Vector Spaces

4.1 Vector Spaces

The preceding chapters dealt primarily with coordinate vectors and matrices. We now isolate the algebraic structure that these examples have in common.

Definition 4.1.1. Let K be a field. A vector space over K is a set V equipped with two operations:

$$V \times V \to V, \qquad (u, v) \mapsto u + v,$$

called vector addition, and

$$K \times V \to V, \qquad (a, v) \mapsto av,$$

called scalar multiplication, satisfying the following properties for all $u, v, w \in V$ and $a, b \in K$:

1. $$(u + v) + w = u + (v + w);$$
2. $$u + v = v + u;$$
3. there exists $0 \in V$ such that $v + 0 = v$;
4. for every $v \in V$, there exists $-v \in V$ such that $v + (-v) = 0$;
5. $$1v = v;$$
6. $$a(bv) = (ab)v;$$
7. $$a(u + v) = au + av;$$
8. $$(a + b)v = av + bv.$$

The elements of V are called vectors, while the elements of K are called scalars.

Example 4.1.2. The space

$$K^n = \{(x_1, \ldots, x_n) : x_i \in K\}$$

is a vector space over K under componentwise addition and scalar multiplication.

Example 4.1.3. The set $M_{m\times n}(K)$ of all $m \times n$ matrices over K is a vector space over K.

Example 4.1.4. The set

$$K[x]_{\leq n} = \{a_0 + a_1x + \cdots + a_nx^n : a_i \in K\}$$

is a vector space over K.

Remark 4.1.5. The objects in a vector space need not be geometric arrows. They may be coordinate vectors, matrices, polynomials, or other objects, provided that addition and scalar multiplication satisfy the vector space axioms.

Several basic identities follow from the axioms:

$$0v = 0, \qquad a0 = 0, \qquad (-1)v = -v.$$

Proposition 4.1.6. The zero vector and the additive inverse of each vector are unique.

4.2 Subspaces

Definition 4.2.1. Let V be a vector space over K. A subset $U \subseteq V$ is a subspace of V if U is itself a vector space under the operations inherited from V.

The subspace criterion gives a more convenient test.

Theorem 4.2.2 (Subspace Criterion)**.** A nonempty subset $U \subseteq V$ is a subspace if and only if for every $u, v \in U$ and $a, b \in K$,

$$au + bv \in U.$$

Proof. If U is a subspace, closure under linear combinations follows immediately. Conversely, suppose $U \neq \varnothing$ and $au + bv \in U$ for all $u, v \in U$ and $a, b \in K$. Choosing $a = b = 0$ gives $0 \in U$. Choosing $a = b = 1$ gives closure under addition, and choosing $a = -1$, $b = 0$ gives additive inverses. The remaining vector space axioms are inherited from V. □

The trivial subspaces are $\{0\}$ and V itself.

Example 4.2.3. For a matrix $A \in M_{m\times n}(K)$, the solution set

$$N(A) = \{x \in K^n : Ax = 0\}$$

of the homogeneous system $Ax = 0$ is a subspace of K^n.

Indeed, if $Ax = 0$ and $Ay = 0$, then for $a, b \in K$,

$$A(ax + by) = aAx + bAy = 0.$$

Thus the homogeneous solution sets studied earlier are naturally vector spaces.

Proposition 4.2.4. The intersection of any collection of subspaces of V is again a subspace of V.

In contrast, the union of two subspaces need not be a subspace.

4.3 Linear Combinations and Span

Definition 4.3.1. Let $v_1, \ldots, v_r \in V$. A vector of the form

$$a_1v_1 + \cdots + a_rv_r, \qquad a_1, \ldots, a_r \in K,$$

is called a linear combination of $v_1, \ldots, v_r$.

Definition 4.3.2. For a subset $S \subseteq V$, the span of S, denoted by $\operatorname{span}(S)$, is the set of all finite linear combinations of vectors in S.

If

$$S = \{v_1, \ldots, v_r\},$$

then

$$\operatorname{span}(v_1, \ldots, v_r) = \{a_1v_1 + \cdots + a_rv_r : a_i \in K\}.$$

Proposition 4.3.3. The span of S is a subspace of V.

Moreover, span(S) is the smallest subspace of V containing S.
That is, if U is a subspace and $S \subseteq U$, then

$$\text{span}(S) \subseteq U.$$

Definition 4.3.4. A subset $S \subseteq V$ is called a spanning set of V if

$$\text{span}(S) = V.$$

Example 4.3.5. The standard vectors

$$e_1 = (1, 0, \ldots, 0), \quad \ldots, \quad e_n = (0, \ldots, 0, 1)$$

span K^n, since

$$(x_1, \ldots, x_n) = x_1 e_1 + \cdots + x_n e_n.$$

4.4 Linear Independence

Definition 4.4.1. Vectors $v_1, \ldots, v_r \in V$ are linearly independent if

$$a_1 v_1 + \cdots + a_r v_r = 0$$

implies

$$a_1 = \cdots = a_r = 0.$$

Otherwise they are linearly dependent.

A set $S \subseteq V$ is linearly independent if every finite collection of distinct vectors from S is linearly independent.

Proposition 4.4.2. A collection $v_1, \ldots, v_r$ is linearly dependent if and only if one of the vectors is a linear combination of the others.

More precisely, if

$$a_1 v_1 + \cdots + a_r v_r = 0$$

with some $a_j \neq 0$, then

$$v_j = -\sum_{i \neq j} \frac{a_i}{a_j} v_i.$$

Corollary 4.4.3. If $v_1, \ldots, v_r$ are linearly independent, then every subset of $\{v_1, \ldots, v_r\}$ is linearly independent.

Proposition 4.4.4. Let $v_1, \ldots, v_r \in K^n$, and form the matrix

$$A = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_r \\ | & & | \end{pmatrix}.$$

Then $v_1, \ldots, v_r$ are linearly independent if and only if the homogeneous system

$$Ax = 0$$

has only the trivial solution.

Thus linear independence generalizes the notion of independent columns encountered in matrix theory.

4.5 Bases

Definition 4.5.1. A basis of a vector space V is a linearly independent spanning set of V.

If

$$\mathcal{B} = (v_1, \ldots, v_n)$$

is a basis of V, then every $v \in V$ can be expressed as

$$v = a_1 v_1 + \cdots + a_n v_n$$

for some scalars $a_1, \ldots, a_n$.
The representation is unique.

Theorem 4.5.2. Let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis of V. Then every $v \in V$ has a unique expression

$$v = a_1 v_1 + \cdots + a_n v_n.$$

Proof. Existence follows because $\mathcal{B}$ spans V.
Suppose

$$v = a_1 v_1 + \cdots + a_n v_n = b_1 v_1 + \cdots + b_n v_n.$$

Then

$$(a_1 - b_1)v_1 + \cdots + (a_n - b_n)v_n = 0.$$

Since $v_1, \ldots, v_n$ are linearly independent,

$$a_i - b_i = 0$$

for every i. Hence $a_i = b_i$. □

The standard basis of K^n is

$$(e_1, \ldots, e_n).$$

The standard basis of $K[x]_{\leq n}$ is

$$(1, x, \ldots, x^n).$$

The standard basis of $M_{m\times n}(K)$ is

$$\{E_{ij} : 1 \leq i \leq m,\ 1 \leq j \leq n\},$$

where E_{ij} has 1 in position (i, j) and 0 elsewhere.

4.6 Basis Extension and Reduction

A central principle of finite-dimensional linear algebra is that linearly independent sets can be enlarged to bases, while spanning sets can be reduced to bases.

Theorem 4.6.1 (Basis Extension Theorem)**.** Let V be a finite-dimensional vector space. Every linearly independent set

$$v_1, \ldots, v_r$$

can be extended to a basis

$$v_1, \ldots, v_r, v_{r+1}, \ldots, v_n$$

of V.

Theorem 4.6.2. Every finite spanning set of V contains a basis of V.

These two results express complementary ideas:

$$\boxed{\text{independent set} \longrightarrow \text{add vectors until it becomes a basis,}}$$

while

$$\boxed{\text{spanning set} \longrightarrow \text{remove redundant vectors until it becomes a basis.}}$$

Corollary 4.6.3. If V is spanned by n vectors, then every linearly independent set in V contains at most n vectors.

4.7 Dimension

Theorem 4.7.1. Any two finite bases of the same vector space contain the same number of vectors.

Definition 4.7.2. If V has a basis containing n vectors, then the dimension of V is

$$\dim V = n.$$

The zero vector space has dimension 0.
Standard examples are

$$\dim K^n = n,$$

$$\dim M_{m\times n}(K) = mn,$$

and

$$\dim K[x]_{\leq n} = n + 1.$$

Theorem 4.7.3. Let V be an n-dimensional vector space.

1. Every linearly independent set in V contains at most n vectors.
2. Every spanning set of V contains at least n vectors.
3. Any linearly independent set of exactly n vectors is a basis.
4. Any spanning set of exactly n vectors is a basis.

Proposition 4.7.4. If $U \subseteq V$ is a subspace and V is finite-dimensional, then

$$\dim U \leq \dim V.$$

Moreover,

$$\dim U = \dim V \iff U = V.$$

This is one of the most useful dimension arguments in finite-dimensional linear algebra.

4.8 Coordinates with Respect to a Basis

Let

$$\mathcal{B} = (v_1, \dots, v_n)$$

be an ordered basis of V. For every $v \in V$, there exist unique scalars $a_1, \dots, a_n$ such that

$$v = a_1 v_1 + \cdots + a_n v_n.$$

Definition 4.8.1. The coordinate vector of v with respect to $\mathcal{B}$ is

$$[v]_{\mathcal{B}} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}.$$

Thus choosing a basis allows every abstract vector in an n-dimensional vector space to be represented by a column vector in K^n.

Remark 4.8.2. The vector v and its coordinate vector $[v]_{\mathcal{B}}$ should be distinguished carefully. The vector v is an element of V, while $[v]_{\mathcal{B}}$ depends on the chosen basis.

Example 4.8.3. Let

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Then

$$\mathcal{B} = (v_1, v_2)$$

is a basis of $\mathbb{R}^2$.
For

$$v = \begin{pmatrix} 3 \\ 1 \end{pmatrix},$$

we solve

$$v = a_1 v_1 + a_2 v_2.$$

Since

$$\begin{pmatrix} 3 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

we obtain

$$[v]_{\mathcal{B}} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Coordinates provide the bridge between abstract vector spaces and matrix computation.

4.9 Sums and Intersections of Subspaces

Let $U, W \subseteq V$ be subspaces.

Definition 4.9.1. The sum of U and W is

$$U + W = \{u + w : u \in U,\ w \in W\}.$$

The set $U + W$ is a subspace of V.
It is the smallest subspace containing both U and W.
The intersection

$$U \cap W$$

is also a subspace.

Theorem 4.9.2 (Dimension Formula)**.** If U and W are finite-dimensional subspaces of V, then

$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W).$$

Proof. Let

$$u_1, \ldots, u_r$$

be a basis of $U \cap W$.
Extend it to bases

$$u_1, \ldots, u_r, v_1, \ldots, v_p$$

of U, and

$$u_1, \ldots, u_r, w_1, \ldots, w_q$$

of W.
Then

$$u_1, \ldots, u_r, v_1, \ldots, v_p, w_1, \ldots, w_q$$

is a basis of $U + W$. Hence

$$\dim(U+W) = r + p + q.$$

Since

$$\dim U = r + p, \qquad \dim W = r + q, \qquad \dim(U \cap W) = r,$$

we obtain

$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W).$$

□

Equivalently,

$$\dim(U+W) + \dim(U \cap W) = \dim U + \dim W.$$

4.10 Direct Sums

The sum $U + W$ does not in general give a unique decomposition of a vector.

Definition 4.10.1. The sum $U+W$ is called a direct sum if every vector $v \in U+W$ has a unique representation

$$v = u + w, \qquad u \in U, \quad w \in W.$$

In this case we write

$$U \oplus W.$$

Theorem 4.10.2. For subspaces $U, W \subseteq V$, the following are equivalent:

1. $U + W$ is a direct sum;
2. $$U \cap W = \{0\};$$
3. the zero vector has a unique representation

$$0 = u + w, \qquad u \in U, \quad w \in W.$$

Proof. Suppose $U \cap W = \{0\}$, and

$$u_1 + w_1 = u_2 + w_2.$$

Then

$$u_1 - u_2 = w_2 - w_1.$$

The left-hand side lies in U, while the right-hand side lies in W. Hence both belong to $U \cap W = \{0\}$, so

$$u_1 = u_2, \qquad w_1 = w_2.$$

Thus the representation is unique.
Conversely, if $x \in U \cap W$, then

$$0 = 0 + 0 = x + (-x)$$

are two representations of 0 with components in U and W. Uniqueness implies $x = 0$. □

If $V = U \oplus W$, then

$$\dim V = \dim U + \dim W.$$

More generally, for subspaces $U_1, \ldots, U_r$, we write

$$V = U_1 \oplus \cdots \oplus U_r$$

if every $v \in V$ has a unique representation

$$v = u_1 + \cdots + u_r, \qquad u_i \in U_i.$$

Then

$$\dim V = \dim U_1 + \cdots + \dim U_r.$$

4.11 Complementary Subspaces

Definition 4.11.1. Let U be a subspace of V. A subspace $W \subseteq V$ is called a complementary subspace of U if

$$V = U \oplus W.$$

Theorem 4.11.2. Every subspace of a finite-dimensional vector space has a complementary subspace.

Proof. Let

$$u_1, \ldots, u_r$$

be a basis of U. By the basis extension theorem, extend this to a basis

$$u_1, \ldots, u_r, w_1, \ldots, w_s$$

of V.
Define

$$W = \operatorname{span}(w_1, \ldots, w_s).$$

Then

$$V = U + W.$$

Since the full collection is linearly independent,

$$U \cap W = \{0\}.$$

Therefore

$$V = U \oplus W.$$

□

The complementary subspace is generally not unique.

Example 4.11.3. Let

$$U = \operatorname{span}\left\{\begin{pmatrix}1\\0\end{pmatrix}\right\} \subseteq \mathbb{R}^2.$$

Both

$$W_1 = \operatorname{span}\left\{\begin{pmatrix}0\\1\end{pmatrix}\right\}$$

and

$$W_2 = \operatorname{span}\left\{\begin{pmatrix}1\\1\end{pmatrix}\right\}$$

satisfy

$$\mathbb{R}^2 = U \oplus W_i.$$

4.12 Direct Products of Vector Spaces

Let $V_1, \ldots, V_r$ be vector spaces over the same field K. Their Cartesian product

$$V_1 \times \cdots \times V_r$$

becomes a vector space under componentwise operations:

$$(v_1, \ldots, v_r) + (w_1, \ldots, w_r) = (v_1 + w_1, \ldots, v_r + w_r),$$

and

$$a(v_1, \ldots, v_r) = (av_1, \ldots, av_r).$$

This vector space is called the direct product of $V_1, \ldots, V_r$.
If all spaces are finite-dimensional, then

$$\dim(V_1 \times \cdots \times V_r) = \dim V_1 + \cdots + \dim V_r.$$

Inside the product, define

$$\widetilde{V}_i = \{(0, \ldots, 0, v_i, 0, \ldots, 0) : v_i \in V_i\}.$$

Then

$$V_1 \times \cdots \times V_r = \widetilde{V}_1 \oplus \cdots \oplus \widetilde{V}_r.$$

This gives a concrete model of a direct sum decomposition.

4.13 Matrix Spaces Revisited

The abstract theory now clarifies several notions introduced earlier.
For $A \in M_{m\times n}(K)$, let its columns be

$$c_1(A), \dots, c_n(A) \in K^m.$$

The column space is

$$C(A) = \operatorname{span}\{c_1(A), \dots, c_n(A)\}.$$

Similarly, the row space is

$$R(A) = \operatorname{span}\{r_1(A), \dots, r_m(A)\}.$$

The null space is

$$N(A) = \{x \in K^n : Ax = 0\}.$$

All three are vector spaces.
The rank theorem for matrices may now be interpreted as

$$\operatorname{rank}(A) = \dim C(A) = \dim R(A).$$

Thus rank is the dimension of the space generated by the columns, and equally the dimension of the space generated by the rows.
If A has n columns, then the number of free variables in the homogeneous system $Ax = 0$ is

$$n - \operatorname{rank}(A).$$

Hence

$$\dim N(A) = n - \operatorname{rank}(A).$$

This is the matrix form of the rank–nullity principle that will be developed systematically for linear maps in the next chapter.

4.14 From Vector Spaces to Linear Maps

Vector spaces provide the abstract setting in which linear algebra takes place. The central concepts developed in this chapter are connected by

$$\boxed{\text{linear combinations} \longrightarrow \text{span and independence} \longrightarrow \text{bases} \longrightarrow \text{dimension.}}$$

Subspaces describe linear structures contained inside larger vector spaces, while direct sums describe how a vector space can be decomposed into simpler pieces.
A basis also explains why finite-dimensional vector spaces can be studied using matrices: once a basis is chosen, every vector is represented by a coordinate vector in K^n.
The next question is therefore natural. Given two vector spaces V and W, which maps

$$T : V \to W$$

respect their linear structure?
These maps are the linear maps studied in the next chapter. They will provide the conceptual interpretation of matrix multiplication, kernel, image, rank, change of basis, and matrix similarity.

Chapter 5

Linear Maps

5.1 Linear Maps

Let V and W be vector spaces over the same field K.

Definition 5.1.1. A map

$$T : V \to W$$

is called a linear map if for every $u, v \in V$ and every $a, b \in K$,

$$T(au + bv) = aT(u) + bT(v).$$

Equivalently, T is linear if

$$T(u + v) = T(u) + T(v)$$

and

$$T(av) = aT(v)$$

for all $u, v \in V$ and $a \in K$.
Every linear map satisfies

$$T(0) = 0$$

and

$$T(-v) = -T(v).$$

More generally,

$$T\left(\sum_{i=1}^{r} a_i v_i\right) = \sum_{i=1}^{r} a_i T(v_i).$$

Thus a linear map preserves linear combinations.

Example 5.1.2. For a fixed matrix $A \in M_{m\times n}(K)$, the map

$$T : K^n \to K^m, \qquad T(x) = Ax,$$

is linear.

Indeed,

$$T(ax + by) = A(ax + by) = aAx + bAy = aT(x) + bT(y).$$

Example 5.1.3. The differentiation map

$$D : K[x]_{\leq n} \to K[x]_{\leq n-1}, \qquad D(p) = p',$$

is linear.

Example 5.1.4. The trace map

$$\mathrm{tr} : M_n(K) \to K, \qquad \mathrm{tr}(A) = \sum_{i=1}^{n} a_{ii},$$

is linear.

5.2 Linear Operators and Basic Examples

Definition 5.2.1. A linear map from a vector space to itself,

$$T : V \to V,$$

is called a linear operator on V.

The identity map

$$I_V : V \to V, \qquad I_V(v) = v,$$

is a linear operator.
The zero map

$$0 : V \to W, \qquad 0(v) = 0,$$

is linear.
If $S, T : V \to W$ are linear maps and $a, b \in K$, define

$$(aS + bT)(v) = aS(v) + bT(v).$$

Then $aS + bT$ is again linear.
Thus the collection of linear maps

$$\mathrm{Hom}_K(V, W) = \{T : V \to W : T \text{ is linear}\}$$

itself carries a natural vector space structure.
If $V = W$, we write

$$\mathrm{End}_K(V) = \mathrm{Hom}_K(V, V).$$

Remark 5.2.2. At this stage, $\mathrm{Hom}_K(V, W)$ is used only as notation for the vector space of linear maps. More abstract algebraic interpretations are not needed in this volume.

5.3 Kernel and Image

Two subspaces associated with a linear map are fundamental.

Definition 5.3.1. Let $T : V \to W$ be linear. The kernel of T is

$$\ker T = \{v \in V : T(v) = 0\}.$$

Definition 5.3.2. The image of T is

$$\mathrm{Im}\, T = \{T(v) : v \in V\}.$$

Theorem 5.3.3. Both $\ker T$ and $\mathrm{Im}\, T$ are subspaces.

Proof. Let $u, v \in \ker T$ and $a, b \in K$. Then

$$T(au + bv) = aT(u) + bT(v) = 0,$$

so $au + bv \in \ker T$. Hence $\ker T$ is a subspace.
Now let $y_1, y_2 \in \mathrm{Im}\, T$. Then

$$y_1 = T(v_1), \qquad y_2 = T(v_2)$$

for some $v_1, v_2 \in V$. Therefore

$$ay_1 + by_2 = aT(v_1) + bT(v_2) = T(av_1 + bv_2),$$

so $ay_1 + by_2 \in \mathrm{Im}\, T$. □

The kernel measures how much information is lost by T, while the image measures which vectors in W can actually be reached.

Theorem 5.3.4. A linear map $T : V \to W$ is injective if and only if

$$\ker T = \{0\}.$$

Proof. Suppose T is injective. If $v \in \ker T$, then

$$T(v) = 0 = T(0).$$

Injectivity gives $v = 0$.
Conversely, suppose $\ker T = \{0\}$ and

$$T(u) = T(v).$$

Then

$$T(u - v) = 0,$$

so $u - v \in \ker T$. Hence $u = v$. □

Similarly,

$$T \text{ is surjective } \iff \operatorname{Im} T = W.$$

5.4 Isomorphisms

Definition 5.4.1. A bijective linear map

$$T : V \to W$$

is called an isomorphism.

If an isomorphism exists, we write

$$V \cong W.$$

Theorem 5.4.2. If $T : V \to W$ is a bijective linear map, then its inverse

$$T^{-1} : W \to V$$

is also linear.

Proof. Let $y_1, y_2 \in W$. Since T is surjective, write

$$y_1 = T(v_1), \qquad y_2 = T(v_2).$$

Then

$$T^{-1}(ay_1 + by_2) = T^{-1}\Big(T(av_1 + bv_2)\Big) = av_1 + bv_2.$$

Therefore

$$T^{-1}(ay_1 + by_2) = aT^{-1}(y_1) + bT^{-1}(y_2).$$

□

Theorem 5.4.3. Two finite-dimensional vector spaces over K are isomorphic if and only if they have the same dimension.

Proof. Suppose

$$\dim V = \dim W = n.$$

Choose bases

$$(v_1, \ldots, v_n)$$

of V and

$$(w_1, \ldots, w_n)$$

of W. Define

$$T(v_i) = w_i$$

and extend linearly. Then T is an isomorphism.
Conversely, an isomorphism maps a basis of V to a basis of W, so the dimensions are equal. □

In particular, every n-dimensional vector space over K is isomorphic to K^n:

$$V \cong K^n.$$

5.5 A Linear Map Is Determined by a Basis

A major advantage of bases is that a linear map is completely determined by its values on basis vectors.

Theorem 5.5.1. Let

$$\mathcal{B} = (v_1, \ldots, v_n)$$

be a basis of V. Given arbitrary vectors

$$w_1, \ldots, w_n \in W,$$

there exists a unique linear map

$$T : V \to W$$

such that

$$T(v_i) = w_i$$

for $i = 1, \ldots, n$.

Proof. Every $v \in V$ has a unique representation

$$v = a_1 v_1 + \cdots + a_n v_n.$$

Define

$$T(v) = a_1 w_1 + \cdots + a_n w_n.$$

This is well defined because the coefficients a_i are unique, and it is immediately linear. Any linear map satisfying $T(v_i) = w_i$ must satisfy

$$T(v) = a_1 T(v_1) + \cdots + a_n T(v_n) = a_1 w_1 + \cdots + a_n w_n,$$

so the map is unique. □

Thus specifying a linear map amounts to specifying the images of a basis.

5.6 Rank and Nullity

Definition 5.6.1. Let $T : V \to W$ be linear, with V finite-dimensional. The rank of T is

$$\operatorname{rank}(T) = \dim(\operatorname{Im} T).$$

The nullity of T is

$$\operatorname{nullity}(T) = \dim(\ker T).$$

Theorem 5.6.2 (Rank–Nullity Theorem)**.** Let $T : V \to W$ be a linear map with V finite-dimensional. Then

$$\dim V = \operatorname{rank}(T) + \operatorname{nullity}(T).$$

Equivalently,

$$\dim V = \dim(\ker T) + \dim(\operatorname{Im} T).$$

Proof. Let

$$v_1, \ldots, v_r$$

be a basis of $\ker T$. Extend it to a basis

$$v_1, \ldots, v_r, v_{r+1}, \ldots, v_n$$

of V.
We claim that

$$T(v_{r+1}), \ldots, T(v_n)$$

is a basis of $\operatorname{Im} T$.
First, if $y \in \operatorname{Im} T$, then $y = T(v)$ for some

$$v = \sum_{i=1}^{n} a_i v_i.$$

Since $T(v_i) = 0$ for $i \leq r$,

$$y = \sum_{i=r+1}^{n} a_i T(v_i),$$

so these vectors span $\operatorname{Im} T$.
Now suppose

$$\sum_{i=r+1}^{n} a_i T(v_i) = 0.$$

Then

$$T\left(\sum_{i=r+1}^{n} a_i v_i\right) = 0,$$

so

$$\sum_{i=r+1}^{n} a_i v_i \in \ker T = \operatorname{span}(v_1, \ldots, v_r).$$

Since $v_1, \ldots, v_n$ are linearly independent, all $a_i = 0$. Hence the images are linearly independent.
Therefore

$$\operatorname{rank}(T) = n - r, \qquad \operatorname{nullity}(T) = r,$$

and the result follows. □

Corollary 5.6.3. If V and W are finite-dimensional and

$$\dim V > \dim W,$$

then every linear map $T : V \to W$ has a nontrivial kernel.

Indeed,

$$\operatorname{rank}(T) \leq \dim W < \dim V,$$

so

$$\operatorname{nullity}(T) > 0.$$

Corollary 5.6.4. If

$$\dim V = \dim W < \infty,$$

then for a linear map $T : V \to W$, the following are equivalent:

1. T is injective;
2. T is surjective;
3. T is bijective;
4. T is an isomorphism.

5.7 Matrix Representation of a Linear Map

We now connect abstract linear maps with matrices.
Let

$$T : V \to W$$

be linear. Suppose

$$\mathcal{B} = (v_1, \ldots, v_n)$$

is an ordered basis of V, and

$$\mathcal{C} = (w_1, \ldots, w_m)$$

is an ordered basis of W.
For each j,

$$T(v_j) = a_{1j}w_1 + \cdots + a_{mj}w_m.$$

Definition 5.7.1. The matrix of T with respect to the bases $\mathcal{B}$ and $\mathcal{C}$ is

$$[T]_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}.$$

Its j-th column is precisely

$$[T(v_j)]_{\mathcal{C}}.$$

Therefore

$$[T]_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \cdots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix}.$$

For every $v \in V$,

$$[T(v)]_{\mathcal{C}} = [T]_{\mathcal{C}\leftarrow\mathcal{B}}[v]_{\mathcal{B}}.$$

This identity is one of the fundamental formulas of linear algebra.

Remark 5.7.2. The linear map T is intrinsic. Its matrix representation depends on the chosen bases $\mathcal{B}$ and $\mathcal{C}$.

5.8 Matrices and Linear Maps between Coordinate Spaces

Let

$$A \in M_{m\times n}(K).$$

Then

$$T_A : K^n \to K^m, \qquad T_A(x) = Ax,$$

is linear.
Conversely, every linear map

$$T : K^n \to K^m$$

is of this form for a unique matrix A.
Indeed, let $e_1, \dots, e_n$ be the standard basis of K^n. Since T is determined by

$$T(e_1), \dots, T(e_n),$$

define

$$A = \begin{pmatrix} | & & | \\ T(e_1) & \cdots & T(e_n) \\ | & & | \end{pmatrix}.$$

Then

$$T(x) = Ax.$$

Thus there is a natural correspondence

$$\boxed{\mathrm{Hom}_K(K^n, K^m) \longleftrightarrow M_{m\times n}(K).}$$

More generally, after choosing bases of finite-dimensional vector spaces V and W,

$$\mathrm{Hom}_K(V, W) \longleftrightarrow M_{m\times n}(K),$$

where

$$n = \dim V, \qquad m = \dim W.$$

5.9 Kernel, Image, and Rank of a Matrix Representation

Let $T : V \to W$ be linear and let

$$A = [T]_{\mathcal{C}\leftarrow\mathcal{B}}.$$

Then

$$[T(v)]_{\mathcal{C}} = A[v]_{\mathcal{B}}.$$

Hence

$$v \in \ker T \iff A[v]_{\mathcal{B}} = 0.$$

Similarly, the coordinate vectors of elements of $\mathrm{Im}\, T$ are exactly the vectors in the column space of A.
Therefore

$$\mathrm{rank}(T) = \mathrm{rank}(A).$$

Also,

$$\mathrm{nullity}(T) = \dim N(A).$$

Thus the matrix identity

$$\mathrm{rank}(A) + \dim N(A) = n$$

is precisely the rank–nullity theorem expressed in coordinates.

5.10 Composition and Matrix Multiplication

Suppose

$$T : U \to V, \qquad S : V \to W$$

are linear maps.
Then their composition

$$S \circ T : U \to W$$

is linear.
Choose ordered bases

$$\mathcal{A}$$

of U,

$$\mathcal{B}$$

of V, and

$$\mathcal{C}$$

of W.
Then

$$[T(u)]_{\mathcal{B}} = [T]_{\mathcal{B}\leftarrow\mathcal{A}}[u]_{\mathcal{A}},$$

and

$$[S(v)]_{\mathcal{C}} = [S]_{\mathcal{C}\leftarrow\mathcal{B}}[v]_{\mathcal{B}}.$$

Therefore

$$[(S \circ T)(u)]_{\mathcal{C}} = [S]_{\mathcal{C}\leftarrow\mathcal{B}}[T]_{\mathcal{B}\leftarrow\mathcal{A}}[u]_{\mathcal{A}}.$$

Hence

$$\boxed{[S \circ T]_{\mathcal{C}\leftarrow\mathcal{A}} = [S]_{\mathcal{C}\leftarrow\mathcal{B}}[T]_{\mathcal{B}\leftarrow\mathcal{A}}.}$$

Thus matrix multiplication represents composition of linear maps.
This explains both the definition and the order of matrix multiplication.

5.11 Invertible Linear Maps and Inverse Matrices

Let $T : V \to W$ be an isomorphism. Then $T^{-1} : W \to V$ is linear.
Choose bases $\mathcal{B}$ of V and $\mathcal{C}$ of W. Since

$$T^{-1} \circ T = I_V,$$

we obtain

$$[T^{-1}]_{\mathcal{B}\leftarrow\mathcal{C}}[T]_{\mathcal{C}\leftarrow\mathcal{B}} = I.$$

Similarly,

$$[T]_{\mathcal{C}\leftarrow\mathcal{B}}[T^{-1}]_{\mathcal{B}\leftarrow\mathcal{C}} = I.$$

Therefore

$$[T^{-1}]_{\mathcal{B}\leftarrow\mathcal{C}} = [T]^{-1}_{\mathcal{C}\leftarrow\mathcal{B}}.$$

Thus

$$\boxed{\text{invertible linear map} \longleftrightarrow \text{invertible matrix.}}$$

This gives the conceptual explanation for the inverse matrices studied in Chapter 3.

5.12 Change of Basis

Let V be an n-dimensional vector space, and let

$$\mathcal{B} = (v_1, \dots, v_n), \qquad \mathcal{C} = (w_1, \dots, w_n)$$

be two ordered bases.

Definition 5.12.1. The change-of-basis matrix from $\mathcal{B}$-coordinates to $\mathcal{C}$-coordinates is the unique invertible matrix P such that

$$[v]_{\mathcal{C}} = P[v]_{\mathcal{B}}$$

for every $v \in V$.

Its columns are

$$P = \begin{pmatrix} | & & | \\ [v_1]_{\mathcal{C}} & \cdots & [v_n]_{\mathcal{C}} \\ | & & | \end{pmatrix}.$$

Since both $\mathcal{B}$ and $\mathcal{C}$ are bases, P is invertible.
The reverse coordinate change is

$$[v]_{\mathcal{B}} = P^{-1}[v]_{\mathcal{C}}.$$

Remark 5.12.2. Care is required with the direction of a change-of-basis matrix. The formula

$$[v]_{\mathcal{C}} = P[v]_{\mathcal{B}}$$

should always be used to determine what P means.

5.13 Change of Matrix Representation

Let

$$T : V \to W$$

be linear.
Suppose $\mathcal{B}, \mathcal{B}'$ are bases of V, while $\mathcal{C}, \mathcal{C}'$ are bases of W.
Let P satisfy

$$[v]_{\mathcal{B}} = P[v]_{\mathcal{B}'}$$

and let Q satisfy

$$[w]_{\mathcal{C}} = Q[w]_{\mathcal{C}'}.$$

Then

$$[T(v)]_{\mathcal{C}} = [T]_{\mathcal{C}\leftarrow\mathcal{B}}[v]_{\mathcal{B}}.$$

Substituting the coordinate changes gives

$$Q[T(v)]_{\mathcal{C}'} = [T]_{\mathcal{C}\leftarrow\mathcal{B}}P[v]_{\mathcal{B}'}.$$

Hence

$$[T(v)]_{\mathcal{C}'} = Q^{-1}[T]_{\mathcal{C}\leftarrow\mathcal{B}}P[v]_{\mathcal{B}'}.$$

Therefore

$$\boxed{[T]_{\mathcal{C}'\leftarrow\mathcal{B}'} = Q^{-1}[T]_{\mathcal{C}\leftarrow\mathcal{B}}P.}$$

This is the general change-of-basis formula for the matrix of a linear map.

5.14 Similarity

The most important special case occurs for a linear operator

$$T : V \to V.$$

Let $\mathcal{B}$ and $\mathcal{C}$ be two bases of V, and let P satisfy

$$[v]_{\mathcal{B}} = P[v]_{\mathcal{C}}.$$

If

$$A = [T]_{\mathcal{B}}, \qquad B = [T]_{\mathcal{C}},$$

then

$$\boxed{B = P^{-1}AP.}$$

Definition 5.14.1. Two square matrices $A, B \in M_n(K)$ are similar if there exists an invertible matrix P such that

$$B = P^{-1}AP.$$

Similarity is therefore not an arbitrary matrix operation. Similar matrices are different coordinate representations of the same linear operator.

Proposition 5.14.2. Similarity is an equivalence relation on $M_n(K)$.

Indeed:

$$A = I^{-1}AI,$$

so similarity is reflexive.
If

$$B = P^{-1}AP,$$

then

$$A = PBP^{-1},$$

so similarity is symmetric.
If

$$B = P^{-1}AP$$

and

$$C = Q^{-1}BQ,$$

then

$$C = Q^{-1}P^{-1}APQ = (PQ)^{-1}A(PQ),$$

so similarity is transitive.

5.15 Similarity Invariants

Because similar matrices represent the same linear operator, many important quantities are unchanged under similarity.
For example, if

$$B = P^{-1}AP,$$

then

$$\operatorname{rank}(B) = \operatorname{rank}(A).$$

Also,

$$B^k = P^{-1}A^kP$$

for every $k \geq 0$.
More generally, for any polynomial p,

$$p(B) = P^{-1}p(A)P.$$

Consequently,

$$p(A) = 0 \iff p(B) = 0.$$

Later, after determinants and characteristic polynomials have been developed, we will prove that similar matrices also have the same determinant, trace, characteristic polynomial, eigenvalues, and minimal polynomial.
Thus similarity leads naturally to a central question:

Which properties of a matrix depend only on the underlying linear operator?

The search for such invariants is one of the main themes of the remaining chapters.

5.16 Invariant Subspaces

Let $T : V \to V$ be a linear operator.

Definition 5.16.1. A subspace $U \subseteq V$ is called T-invariant if

$$T(U) \subseteq U.$$

Equivalently, whenever $u \in U$,

$$T(u) \in U.$$

The trivial subspaces

$$\{0\} \qquad \text{and} \qquad V$$

are always invariant.

Example 5.16.2. Let

$$T : \mathbb{R}^2 \to \mathbb{R}^2$$

be defined by

$$T(x, y) = (2x, 3y).$$

Then

$$U_1 = \operatorname{span}\left\{\begin{pmatrix}1\\0\end{pmatrix}\right\}$$

and

$$U_2 = \operatorname{span}\left\{\begin{pmatrix}0\\1\end{pmatrix}\right\}$$

are both T-invariant.

If

$$V = U \oplus W$$

and both U and W are T-invariant, then choosing bases of U and W produces a block diagonal representation

$$[T] = \begin{pmatrix}[T|_U] & 0\\ 0 & [T|_W]\end{pmatrix}.$$

If only U is invariant and the basis of U is extended to a basis of V, the matrix of T has block upper triangular form

$$[T] = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}.$$

Thus invariant subspaces allow a linear operator to be decomposed into simpler pieces.

5.17 One-Dimensional Invariant Subspaces

A particularly important case is a one-dimensional invariant subspace.
Let

$$U = \operatorname{span}(v), \qquad v \neq 0.$$

Then U is T-invariant if and only if

$$T(v) \in \operatorname{span}(v).$$

Therefore there exists $\lambda \in K$ such that

$$T(v) = \lambda v.$$

This is precisely the eigenvector equation.
Thus eigenvectors can be interpreted geometrically as generators of one-dimensional invariant subspaces.
The systematic study of eigenvalues and eigenvectors will be developed after determinants and polynomial methods have been introduced.

5.18 The Structural Role of Linear Maps

The main objects of the preceding chapters are now connected:

$$\boxed{\text{vector spaces} \longrightarrow \text{linear maps} \longrightarrow \text{matrices after bases are chosen.}}$$

Kernel and image describe the internal structure of a linear map, and the rank–nullity theorem gives

$$\dim V = \dim \ker T + \dim \operatorname{Im} T.$$

Matrix multiplication represents composition:

$$[S \circ T] = [S][T].$$

Invertible matrices represent isomorphisms:

$$[T^{-1}] = [T]^{-1}.$$

Most importantly, changing the basis of a linear operator changes its matrix by similarity:

$$A \longmapsto P^{-1}AP.$$

From this point onward, the goal is no longer merely to calculate with a particular matrix. We seek information that survives changes of coordinates.
The next chapter introduces determinants. Determinants provide one of the first major invariants of square matrices and give a powerful criterion for invertibility. They will also lead to characteristic polynomials and, eventually, to eigenvalue theory.

Chapter 6

Determinants

6.1 Geometric Motivation and the Determinant

For a real 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the columns

$$v_1 = \begin{pmatrix} a \\ c \end{pmatrix}, \qquad v_2 = \begin{pmatrix} b \\ d \end{pmatrix}$$

determine a parallelogram in $\mathbb{R}^2$.
Its signed area is

$$ad - bc.$$

This motivates the definition

$$\det A = ad - bc.$$

In dimension three, if the columns of A are $v_1, v_2, v_3 \in \mathbb{R}^3$, then

$$|\det A|$$

is the volume of the parallelepiped generated by these vectors.
The sign records orientation.
The determinant therefore measures how a square matrix changes oriented volume.
For a linear transformation

$$T : \mathbb{R}^n \to \mathbb{R}^n,$$

the quantity $|\det T|$ represents the factor by which n-dimensional volume is scaled.
If

$$\det T = 0,$$

then T collapses the ambient space into a lower-dimensional subspace.

6.2 Permutations

To define determinants in arbitrary dimension, we first introduce permutations.

Definition 6.2.1. A permutation of $\{1, \dots, n\}$ is a bijection

$$\sigma : \{1, \dots, n\} \to \{1, \dots, n\}.$$

The set of all permutations is denoted by S_n.

Since there are $n!$ possible orderings of n distinct objects,

$$|S_n| = n!.$$

Definition 6.2.2. An inversion of a permutation $\sigma \in S_n$ is a pair (i, j) satisfying

$$i < j$$

but

$$\sigma(i) > \sigma(j).$$

Let $N(\sigma)$ denote the number of inversions of σ.

Definition 6.2.3. The sign of a permutation is

$$\operatorname{sgn}(\sigma) = (-1)^{N(\sigma)}.$$

Thus

$$\operatorname{sgn}(\sigma) = \begin{cases} 1, & \sigma \text{ is even}, \\ -1, & \sigma \text{ is odd}. \end{cases}$$

A transposition exchanges two elements and leaves all others fixed.

Proposition 6.2.4. Every permutation can be written as a product of transpositions. Moreover, the parity of the number of transpositions in such a decomposition is independent of the decomposition.

Hence

$$\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau).$$

6.3 The Leibniz Formula

Let

$$A = (a_{ij}) \in M_n(K).$$

Definition 6.3.1. The determinant of A is

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

Equivalently,

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)}.$$

For $n = 2$,

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

For $n = 3$,

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

equals

$$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.$$

Remark 6.3.2. The Sarrus rule is a convenient mnemonic for 3×3 determinants, but it does not generalize to higher dimensions.

6.4 Multilinearity and Alternation

The determinant is best understood through its structural properties.
Write the rows of A as

$$r_1, \ldots, r_n.$$

Then we may regard the determinant as a function

$$\det : (K^n)^n \to K, \qquad (r_1, \ldots, r_n) \mapsto \det A.$$

Theorem 6.4.1. The determinant is multilinear in the rows. In particular, if the i-th row is replaced by $u + v$, then

$$\det(r_1, \ldots, u + v, \ldots, r_n)$$

equals

$$\det(r_1, \ldots, u, \ldots, r_n) + \det(r_1, \ldots, v, \ldots, r_n).$$

Also,

$$\det(r_1, \ldots, cr_i, \ldots, r_n) = c \det(r_1, \ldots, r_i, \ldots, r_n).$$

Definition 6.4.2. A multilinear map is alternating if its value is zero whenever two arguments are equal.

Theorem 6.4.3. The determinant is alternating.

Hence if two rows of A are equal, then

$$\det A = 0.$$

More generally, interchanging two rows changes the sign:

$$\det B = -\det A.$$

The determinant is normalized by

$$\det I_n = 1.$$

Theorem 6.4.4. The determinant is the unique function

$$D : M_n(K) \to K$$

that is multilinear in the rows, alternating, and satisfies

$$D(I_n) = 1.$$

The same properties hold with columns in place of rows.

6.5 Elementary Row Operations

The determinant behaves predictably under elementary row operations.
If B is obtained from A by interchanging two rows, then

$$\det B = -\det A.$$

If one row is multiplied by $c \in K$, then

$$\det B = c \det A.$$

If a scalar multiple of one row is added to another row, then

$$\det B = \det A.$$

The last property follows from multilinearity and alternation. If

$$r_j \mapsto r_j + cr_i,$$

then

$$\det(r_1, \ldots, r_j + cr_i, \ldots, r_n)$$

equals

$$\det(r_1, \ldots, r_j, \ldots, r_n) + c \det(r_1, \ldots, r_i, \ldots, r_i, \ldots, r_n).$$

The second term vanishes because two rows coincide.
These rules make Gaussian elimination an efficient method for computing determinants.

Example 6.5.1. Suppose row reduction transforms A into an upper triangular matrix

$$U = \begin{pmatrix} u_{11} & * & \cdots & * \\ 0 & u_{22} & \cdots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & u_{nn} \end{pmatrix}$$

using only operations of the form

$$R_j \mapsto R_j + cR_i.$$

Then

$$\det A = \det U = u_{11}u_{22}\cdots u_{nn}.$$

If row interchanges or row scalings are used, the corresponding sign and scalar factors must also be recorded.

6.6 Triangular and Block Triangular Matrices

Theorem 6.6.1. If $A \in M_n(K)$ is upper or lower triangular, then

$$\det A = \prod_{i=1}^{n} a_{ii}.$$

In particular,

$$\det \operatorname{diag}(d_1, \ldots, d_n) = d_1 \cdots d_n.$$

A similar formula holds for block triangular matrices.

Theorem 6.6.2. If

$$A = \begin{pmatrix} B & C \\ 0 & D \end{pmatrix},$$

where B and D are square matrices, then

$$\det A = \det B \det D.$$

Likewise,

$$\det \begin{pmatrix} B & 0 \\ C & D \end{pmatrix} = \det B \det D.$$

In particular,

$$\det \begin{pmatrix} B & 0 \\ 0 & D \end{pmatrix} = \det B \det D.$$

6.7 Transpose and Determinant

Theorem 6.7.1. For every $A \in M_n(K)$,

$$\det(A^T) = \det A.$$

Thus all determinant formulas stated in terms of rows have analogous versions in terms of columns.
For example, if two columns are exchanged, the determinant changes sign, and adding a multiple of one column to another does not change the determinant.

6.8 Multiplicativity of the Determinant

One of the fundamental properties of determinants is compatibility with matrix multiplication.

Theorem 6.8.1. For $A, B \in M_n(K)$,

$$\det(AB) = \det A \det B.$$

Proof. Fix $A \in M_n(K)$ and define

$$D(B) = \det(AB).$$

As a function of the columns of B, the map D is multilinear and alternating. Therefore, by the uniqueness characterization of the determinant,

$$D(B) = D(I_n) \det B.$$

Since

$$D(I_n) = \det A,$$

we obtain

$$\det(AB) = \det A \det B.$$

□

By induction,

$$\det(A_1 A_2 \cdots A_r) = \prod_{i=1}^{r} \det(A_i).$$

For every positive integer m,

$$\det(A^m) = (\det A)^m.$$

6.9 Determinant and Invertibility

The determinant completely detects invertibility.

Theorem 6.9.1. For $A \in M_n(K)$, the following are equivalent:

1. A is invertible;
2. $\det A \neq 0$;
3. $\operatorname{rank}(A) = n$;
4. the columns of A are linearly independent;
5. the rows of A are linearly independent;
6. $Ax = 0$ has only the trivial solution.

Proof. If A is invertible, then

$$AA^{-1} = I_n.$$

Taking determinants,

$$\det A \det A^{-1} = 1,$$

so $\det A \neq 0$.

Conversely, suppose A is not invertible. Then

$$\operatorname{rank}(A) < n,$$

so its rows are linearly dependent. By multilinearity and alternation,

$$\det A = 0.$$

□

Corollary 6.9.2. If A is invertible, then

$$\det(A^{-1}) = \frac{1}{\det A}.$$

This gives another interpretation of the determinant:

$$\boxed{\det A = 0 \iff A \text{ collapses dimension.}}$$

6.10 Minors and Cofactors

Let

$$A = (a_{ij}) \in M_n(K).$$

For $1 \leq i, j \leq n$, let $A(i|j)$ denote the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j.

Definition 6.10.1. The (i, j)-minor of A is

$$M_{ij} = \det A(i|j).$$

Definition 6.10.2. The (i, j)-cofactor of A is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

Thus the signs of the cofactors have the checkerboard pattern

$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

Definition 6.10.3. The cofactor matrix of A is

$$C = (C_{ij}).$$

The adjugate matrix of A is

$$\operatorname{adj}(A) = C^T.$$

Hence

$$\operatorname{adj}(A)_{ij} = C_{ji}.$$

6.11 Laplace Expansion

Theorem 6.11.1 (Laplace Expansion)**.** For any fixed row i,

$$\det A = \sum_{j=1}^{n} a_{ij} C_{ij}.$$

For any fixed column j,

$$\det A = \sum_{i=1}^{n} a_{ij} C_{ij}.$$

Thus the determinant may be expanded along any row or any column.
For a 3×3 matrix,

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix},$$

expansion along the first row gives

$$\det A = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}.$$

Remark 6.11.2. When computing a determinant by Laplace expansion, one should usually choose a row or column containing many zero entries.

6.12 The Adjugate Formula

The cofactors satisfy a stronger identity.

Theorem 6.12.1. For every $A \in M_n(K)$,

$$A \operatorname{adj}(A) = \operatorname{adj}(A) A = (\det A) I_n.$$

Indeed, the diagonal entries are Laplace expansions of $\det A$, while the off-diagonal entries are determinants of matrices with two equal rows or columns and hence vanish.

Corollary 6.12.2. If $\det A \neq 0$, then

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A).$$

Thus the inverse of a nonsingular matrix can be expressed entirely in terms of determinants.
For a 2×2 matrix,

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

if $ad - bc \neq 0$, then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

6.13 Cramer's Rule

Consider the square linear system

$$Ax = b,$$

where

$$A \in M_n(K)$$

and

$$\det A \neq 0.$$

Let A_j denote the matrix obtained from A by replacing its j-th column by b.

Theorem 6.13.1 (Cramer's Rule)**.** The unique solution of $Ax = b$ satisfies

$$x_j = \frac{\det A_j}{\det A}, \qquad j = 1, \ldots, n.$$

Proof. Since A is invertible,

$$x = A^{-1}b.$$

Using the adjugate formula,

$$x = \frac{1}{\det A} \operatorname{adj}(A)b.$$

The j-th component is

$$x_j = \frac{1}{\det A} \sum_{i=1}^{n} C_{ij} b_i.$$

By Laplace expansion, the numerator is precisely

$$\det A_j.$$

Hence

$$x_j = \frac{\det A_j}{\det A}.$$

□

Remark 6.13.2. Cramer's rule is theoretically important because it gives an explicit formula for the solution. For numerical computation, Gaussian elimination is generally far more efficient.

6.14 General Minors

The word minor also has a more general meaning.
Let

$$A \in M_{m \times n}(K).$$

Choose row indices

$$1 \leq i_1 < \cdots < i_r \leq m$$

and column indices

$$1 \leq j_1 < \cdots < j_r \leq n.$$

The corresponding $r \times r$ submatrix is

$$A \begin{pmatrix} i_1, \ldots, i_r \\ j_1, \ldots, j_r \end{pmatrix}.$$

Definition 6.14.1. The determinant of an $r \times r$ submatrix of A is called an r-th order minor of A.

Thus a matrix generally has many minors of the same order.

6.15 Rank and Minors

Determinants give an alternative characterization of matrix rank.

Theorem 6.15.1. Let $A \in M_{m\times n}(K)$. Then

$$\operatorname{rank}(A) = r$$

if and only if:

1. A has at least one nonzero r-th order minor;
2. every $(r+1)$-st order minor of A is zero.

Equivalently,

$$\boxed{\operatorname{rank}(A) = \text{largest order of a nonzero minor of } A.}$$

In particular, for $A \in M_n(K)$,

$$\operatorname{rank}(A) = n \iff \det A \neq 0.$$

This characterization connects the computational notion of rank from Gaussian elimination with determinants of submatrices.

6.16 Determinant under Similarity

Suppose $A, B \in M_n(K)$ are similar:

$$B = P^{-1}AP$$

for some invertible P.
Then

$$\det B = \det(P^{-1}AP).$$

By multiplicativity,

$$\det B = \det(P^{-1}) \det A \det P.$$

Since

$$\det(P^{-1}) = \frac{1}{\det P},$$

we obtain

$$\boxed{\det B = \det A.}$$

Thus determinant is a similarity invariant.
This means that the determinant belongs to the underlying linear operator rather than to a particular matrix representation.

6.17 Determinant of a Linear Operator

Let V be an n-dimensional vector space over K, and let

$$T : V \to V$$

be a linear operator.
Choose an ordered basis $\mathcal{B}$ and let

$$A = [T]_{\mathcal{B}}.$$

Definition 6.17.1. The determinant of T is

$$\det T = \det[T]_{\mathcal{B}}.$$

This definition is independent of the chosen basis.
Indeed, if $\mathcal{C}$ is another basis, then

$$[T]_{\mathcal{C}} = P^{-1}[T]_{\mathcal{B}}P.$$

Therefore

$$\det[T]_{\mathcal{C}} = \det[T]_{\mathcal{B}}.$$

Hence $\det T$ is an intrinsic invariant of the operator.

Theorem 6.17.2. Let $S, T \in \mathrm{End}_K(V)$. Then

$$\det(S \circ T) = \det S \det T.$$

Also,

$$T \text{ is invertible } \iff \det T \neq 0.$$

If T is invertible, then

$$\det(T^{-1}) = (\det T)^{-1}.$$

6.18 Determinant as a Volume Scaling Factor

Over $\mathbb{R}$, the determinant has a geometric interpretation.
Let

$$A \in M_n(\mathbb{R}).$$

The unit cube

$$[0,1]^n$$

is transformed by $x \mapsto Ax$ into the parallelepiped generated by the columns of A.
Its n-dimensional volume is

$$|\det A|\,.$$

Thus:

- if $|\det A| > 1$, volume is expanded;
- if $0 < |\det A| < 1$, volume is contracted;
- if $\det A = 0$, the image has lower dimension;
- if $\det A < 0$, orientation is reversed.

This explains geometrically why

$$\det(AB) = \det A \det B :$$

applying B and then A multiplies volume by the product of the two scaling factors.

6.19 From Determinants to Polynomial Methods

Determinants provide several equivalent descriptions of invertibility:

$$\boxed{A \text{ invertible} \iff \det A \neq 0 \iff \operatorname{rank}(A) = n \iff \ker A = \{0\}.}$$

They also produce an invariant of linear operators:

$$T \longmapsto \det T.$$

The next step is to allow a scalar parameter λ to enter the determinant and study

$$\det(\lambda I - A).$$

This expression is a polynomial in λ, called the characteristic polynomial of A.
The characteristic polynomial will connect determinants with polynomial algebra and eigenvalue problems, providing the main subject of the next chapter.

Chapter 7

Polynomials and Linear Operators

7.1 Polynomial Preliminaries

Let K be a field. A polynomial over K has the form

$$f(X) = a_0 + a_1X + \cdots + a_nX^n, \qquad a_i \in K.$$

If $a_n \neq 0$, then the degree of f is n, written

$$\deg f = n.$$

The coefficient a_n is called the leading coefficient.
A polynomial is monic if its leading coefficient is 1.
The set of all polynomials over K is denoted by $K[X]$.
For every $d \geq 0$, define

$$K[X]_{\leq d} = \{f \in K[X] : \deg f \leq d\} \cup \{0\}.$$

Then $K[X]_{\leq d}$ is a vector space over K with basis

$$1, X, \ldots, X^d,$$

so

$$\dim K[X]_{\leq d} = d + 1.$$

Polynomial addition and multiplication are defined in the usual way. For nonzero polynomials $f, g \in K[X]$,

$$\deg(fg) = \deg f + \deg g.$$

7.2 Polynomial Division

Theorem 7.2.1 (Division Algorithm)**.** Let $f, g \in K[X]$ with $g \neq 0$. Then there exist unique polynomials $q, r \in K[X]$ such that

$$f = qg + r$$

and either $r = 0$ or

$$\deg r < \deg g.$$

The polynomial q is the quotient and r is the remainder.

Example 7.2.2. Dividing

$$f(X) = X^3 + 2X^2 - X + 1$$

by

$$g(X) = X - 1$$

gives

$$f(X) = (X - 1)(X^2 + 3X + 2) + 3.$$

Hence the remainder is 3.

Theorem 7.2.3 (Remainder Theorem)**.** For $f \in K[X]$ and $a \in K$, the remainder obtained when f is divided by $X - a$ is $f(a)$.

Therefore

$$f(X) = (X - a)q(X) + f(a)$$

for some $q \in K[X]$.

Corollary 7.2.4 (Factor Theorem)**.** A scalar $a \in K$ is a root of f if and only if

$$X - a$$

divides f.

7.3 Roots and Multiplicity

Definition 7.3.1. A scalar $a \in K$ is called a root of $f \in K[X]$ if

$$f(a) = 0.$$

Definition 7.3.2. A root a has multiplicity m if

$$(X-a)^m \mid f(X)$$

but

$$(X-a)^{m+1} \nmid f(X).$$

Thus if the distinct roots of f in K are $\lambda_1, \ldots, \lambda_r$, then

$$f(X) = c\prod_{i=1}^{r}(X-\lambda_i)^{m_i} g(X),$$

where g has no roots in K.

Definition 7.3.3. A polynomial $f \in K[X]$ splits over K if it can be written as a product of linear factors:

$$f(X) = c\prod_{i=1}^{r}(X-\lambda_i)^{m_i}, \qquad \lambda_i \in K.$$

Over $\mathbb{C}$, every nonconstant polynomial splits.

Theorem 7.3.4 (Fundamental Theorem of Algebra)**.** Every nonconstant polynomial in $\mathbb{C}[X]$ has a complex root. Equivalently, every polynomial in $\mathbb{C}[X]$ splits completely into linear factors.

Thus every polynomial of degree n over $\mathbb{C}$ has exactly n roots when counted with multiplicity.

7.4 Polynomials of Matrices

Let

$$A \in M_n(K)$$

and let

$$f(X) = a_0 + a_1X + \cdots + a_mX^m.$$

Definition 7.4.1. The matrix obtained by evaluating f at A is

$$f(A) = a_0I_n + a_1A + \cdots + a_mA^m.$$

For example, if

$$f(X) = X^2 - 3X + 2,$$

then

$$f(A) = A^2 - 3A + 2I_n.$$

Polynomial evaluation satisfies

$$(f+g)(A) = f(A) + g(A),$$
$$(fg)(A) = f(A)g(A),$$

and

$$(cf)(A) = cf(A).$$

Since all powers of the same matrix commute,

$$f(A)g(A) = g(A)f(A).$$

Proposition 7.4.2. If

$$B = P^{-1}AP,$$

then for every polynomial $f \in K[X]$,

$$f(B) = P^{-1}f(A)P.$$

Proof. For every $k \geq 0$,

$$B^k = P^{-1}A^kP.$$

Therefore

$$f(B) = \sum_{k=0}^{m} a_k B^k = P^{-1}\left(\sum_{k=0}^{m} a_k A^k\right)P = P^{-1}f(A)P.$$

□

Thus polynomial relations satisfied by a matrix are preserved under similarity.

7.5 Polynomials of Linear Operators

Let

$$T \in \mathrm{End}_K(V).$$

For

$$f(X) = a_0 + a_1X + \cdots + a_mX^m,$$

define

$$f(T) = a_0I_V + a_1T + \cdots + a_mT^m.$$

If $A = [T]_{\mathcal{B}}$, then

$$[f(T)]_{\mathcal{B}} = f(A).$$

Therefore polynomial expressions in matrices are coordinate representations of polynomial expressions in linear operators.
This distinction will be useful in interpreting the Cayley–Hamilton theorem intrinsically.

7.6 Eigenvalues and Eigenvectors

Let

$$T : V \to V$$

be a linear operator.

Definition 7.6.1. A scalar $\lambda \in K$ is an eigenvalue of T if there exists a nonzero vector $v \in V$ such that

$$T(v) = \lambda v.$$

Such a vector v is called an eigenvector associated with λ.

The equation can be rewritten as

$$(T - \lambda I_V)v = 0.$$

Thus λ is an eigenvalue if and only if

$$\ker(T - \lambda I_V) \neq \{0\}.$$

Definition 7.6.2. The eigenspace associated with λ is

$$E_\lambda(T) = \ker(T - \lambda I_V).$$

The eigenspace includes the zero vector, while an eigenvector is by definition nonzero.

Proposition 7.6.3. For every eigenvalue λ, the eigenspace $E_\lambda(T)$ is a nonzero T-invariant subspace.

Indeed, if $v \in E_\lambda(T)$, then

$$T(v) = \lambda v \in E_\lambda(T).$$

Thus the eigenvalue problem is equivalent to finding one-dimensional invariant directions and their higher-dimensional spans.

7.7 Eigenvalues of Matrices

Let

$$A \in M_n(K).$$

A scalar $\lambda \in K$ is an eigenvalue of A if there exists $v \neq 0$ such that

$$Av = \lambda v.$$

Equivalently,

$$(A - \lambda I_n)v = 0$$

has a nontrivial solution.
By the invertibility criterion from Chapter 6,

$$A - \lambda I_n$$

has a nontrivial kernel if and only if

$$\det(A - \lambda I_n) = 0.$$

Since

$$\det(A - \lambda I_n) = (-1)^n \det(\lambda I_n - A),$$

we obtain

$$\boxed{\lambda \text{ is an eigenvalue of } A \iff \det(\lambda I_n - A) = 0.}$$

This converts the eigenvalue problem into a polynomial equation.

7.8 Characteristic Polynomial

Definition 7.8.1. Let

$$A \in M_n(K).$$

The characteristic polynomial of A is

$$\chi_A(X) = \det(XI_n - A).$$

The characteristic polynomial is monic of degree n:

$$\chi_A(X) = X^n + c_{n-1}X^{n-1} + \cdots + c_1X + c_0.$$

A scalar $\lambda \in K$ is an eigenvalue of A precisely when

$$\chi_A(\lambda) = 0.$$

Hence the eigenvalues of A are exactly the roots of its characteristic polynomial that lie in K.

Example 7.8.2. Let

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}.$$

Then

$$\chi_A(X) = \det \begin{pmatrix} X-2 & -1 \\ 0 & X-3 \end{pmatrix} = (X-2)(X-3).$$

Therefore the eigenvalues are 2 and 3.

7.9 Characteristic Polynomial and Similarity

Theorem 7.9.1. Similar matrices have the same characteristic polynomial.

Proof. Suppose

$$B = P^{-1}AP.$$

Then

$$XI_n - B = P^{-1}(XI_n - A)P.$$

Therefore

$$\chi_B(X) = \det(XI_n - B)$$

equals

$$\det(P^{-1}) \det(XI_n - A) \det P = \chi_A(X).$$

□

Thus the characteristic polynomial is a similarity invariant.
Consequently, the characteristic polynomial of a linear operator

$$T : V \to V$$

may be defined by

$$\chi_T(X) = \chi_{[T]_{\mathcal{B}}}(X)$$

for any basis $\mathcal{B}$.
The definition is independent of the chosen basis.

7.10 Trace and the Characteristic Polynomial

For

$$A = (a_{ij}) \in M_n(K),$$

recall that the trace is

$$\operatorname{tr}(A) = \sum_{i=1}^{n} a_{ii}.$$

If

$$\chi_A(X) = X^n + c_{n-1}X^{n-1} + \cdots + c_1 X + c_0,$$

then

$$c_{n-1} = -\operatorname{tr}(A)$$

and

$$c_0 = (-1)^n \det A.$$

Therefore

$$\boxed{\chi_A(X) = X^n - \operatorname{tr}(A)X^{n-1} + \cdots + (-1)^n \det A.}$$

If the characteristic polynomial splits as

$$\chi_A(X) = \prod_{i=1}^{r} (X - \lambda_i)^{m_i},$$

then

$$\operatorname{tr}(A) = \sum_{i=1}^{r} m_i \lambda_i$$

and

$$\det A = \prod_{i=1}^{r} \lambda_i^{m_i}.$$

Thus the trace is the sum of the eigenvalues and the determinant is their product, both counted with algebraic multiplicity.

7.11 Trace as a Similarity Invariant

Theorem 7.11.1. For compatible matrices

$$A \in M_{m\times n}(K), \qquad B \in M_{n\times m}(K),$$

we have

$$\operatorname{tr}(AB) = \operatorname{tr}(BA).$$

Proof. We have

$$\operatorname{tr}(AB) = \sum_{i=1}^{m} (AB)_{ii} = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij} b_{ji}.$$

Interchanging the order of summation gives

$$\operatorname{tr}(AB) = \sum_{j=1}^{n}\sum_{i=1}^{m} b_{ji} a_{ij} = \operatorname{tr}(BA).$$

□

If

$$B = P^{-1}AP,$$

then

$$\operatorname{tr}(B) = \operatorname{tr}(P^{-1}AP) = \operatorname{tr}(APP^{-1}) = \operatorname{tr}(A).$$

Hence trace is also a similarity invariant.
For a linear operator T, define

$$\operatorname{tr}(T) = \operatorname{tr}[T]_{\mathcal{B}},$$

which is independent of the chosen basis.

7.12 Algebraic Multiplicity

Suppose λ is an eigenvalue of A.

Definition 7.12.1. The algebraic multiplicity of λ is its multiplicity as a root of the characteristic polynomial.

Thus if

$$\chi_A(X) = (X - \lambda)^m g(X)$$

and $g(\lambda) \neq 0$, then the algebraic multiplicity of λ is m.
We denote it by

$$m_a(\lambda).$$

Definition 7.12.2. The geometric multiplicity of λ is

$$m_g(\lambda) = \dim E_\lambda(A) = \dim \ker(A - \lambda I_n).$$

The distinction between these two multiplicities is fundamental: algebraic multiplicity comes from the characteristic polynomial, whereas geometric multiplicity measures the number of independent eigenvectors.

Theorem 7.12.3. For every eigenvalue λ,

$$1 \leq m_g(\lambda) \leq m_a(\lambda).$$

Proof. Since λ is an eigenvalue,

$$E_\lambda(A) \neq \{0\},$$

so $m_g(\lambda) \geq 1$.
Let

$$r = m_g(\lambda).$$

Choose a basis

$$v_1, \ldots, v_r$$

of $E_\lambda(A)$ and extend it to a basis of K^n. With respect to this basis, A has block form

$$\begin{pmatrix} \lambda I_r & B \\ 0 & C \end{pmatrix}.$$

Therefore

$$\chi_A(X) = \det \begin{pmatrix} (X - \lambda)I_r & -B \\ 0 & XI - C \end{pmatrix}.$$

Hence

$$\chi_A(X) = (X - \lambda)^r \det(XI - C),$$

so the algebraic multiplicity of λ is at least r. Therefore

$$m_g(\lambda) \leq m_a(\lambda).$$

□

7.13 Eigenvectors for Distinct Eigenvalues

Theorem 7.13.1. Eigenvectors associated with distinct eigenvalues are linearly independent.

Proof. Let

$$v_1, \ldots, v_r$$

be eigenvectors corresponding to distinct eigenvalues

$$\lambda_1, \ldots, \lambda_r.$$

Suppose

$$a_1 v_1 + \cdots + a_r v_r = 0.$$

Assume r is minimal among all nontrivial relations of this form. Applying T gives

$$a_1 \lambda_1 v_1 + \cdots + a_r \lambda_r v_r = 0.$$

Subtracting λ_r times the original relation gives

$$a_1(\lambda_1 - \lambda_r)v_1 + \cdots + a_{r-1}(\lambda_{r-1} - \lambda_r)v_{r-1} = 0.$$

Since the eigenvalues are distinct, every $\lambda_i - \lambda_r \neq 0$. By minimality, all $a_1, \ldots, a_{r-1}$ vanish, and then $a_r = 0$. Hence the vectors are linearly independent. □

Corollary 7.13.2. An $n \times n$ matrix has at most n distinct eigenvalues.

Corollary 7.13.3. If an $n \times n$ matrix has n distinct eigenvalues in K, then it has a basis of eigenvectors.

This already gives an important sufficient condition for diagonalizability, which will be studied systematically in the next chapter.

7.14 Eigenvalues of Triangular Matrices

Let

$$A = \begin{pmatrix} a_{11} & * & \cdots & * \\ 0 & a_{22} & \cdots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn} \end{pmatrix}$$

be upper triangular.
Then

$$XI_n - A$$

is also upper triangular, with diagonal entries

$$X - a_{11}, \ldots, X - a_{nn}.$$

Therefore

$$\chi_A(X) = \prod_{i=1}^{n} (X - a_{ii}).$$

Hence the eigenvalues of a triangular matrix are precisely its diagonal entries, counted with algebraic multiplicity.
The same statement holds for lower triangular matrices.
This fact explains why triangular forms are useful: once a matrix has been triangularized, its eigenvalues can be read directly from the diagonal.

7.15 Cayley–Hamilton Theorem

The characteristic polynomial is not merely a device for finding eigenvalues. A matrix satisfies its own characteristic polynomial.

Theorem 7.15.1 (Cayley–Hamilton Theorem)**.** Let

$$A \in M_n(K)$$

and let

$$\chi_A(X) = \det(XI_n - A).$$

Then

$$\boxed{\chi_A(A) = 0.}$$

The zero on the right-hand side is the zero matrix.
If

$$\chi_A(X) = X^n + c_{n-1}X^{n-1} + \cdots + c_1 X + c_0,$$

then

$$A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I_n = 0.$$

Proof. Let

$$\chi_A(X) = X^n + c_{n-1}X^{n-1} + \cdots + c_0.$$

Over $\mathbb{C}$, the polynomial splits:

$$\chi_A(X) = (X - \lambda_1) \cdots (X - \lambda_n),$$

where the eigenvalues are counted with multiplicity.
A complex matrix is similar to an upper triangular matrix, so choose a basis in which

$$A = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}.$$

Let

$$V_j = \operatorname{span}(v_1, \ldots, v_j), \qquad V_0 = \{0\}.$$

Then

$$(A - \lambda_j I)(V_j) \subseteq V_{j-1}.$$

Therefore

$$(A - \lambda_1 I) \cdots (A - \lambda_n I)(V_n) \subseteq V_0.$$

Hence

$$\chi_A(A) = 0.$$

Since the resulting polynomial identity has coefficients in the original field K, the theorem holds over K. □

Remark 7.15.2. One should not justify the theorem by formally substituting $X = A$ into

$$\chi_A(X) = \det(XI - A).$$

The expression $\det(XI - A)$ is a scalar-valued polynomial in the indeterminate X, while $\chi_A(A)$ is a matrix obtained by polynomial evaluation. The theorem is a genuine result connecting these two constructions.

7.16 Cayley–Hamilton for Linear Operators

Let

$$T \in \mathrm{End}_K(V)$$

with $\dim V = n$.
Define

$$\chi_T(X) = \chi_{[T]_{\mathcal{B}}}(X)$$

for any basis $\mathcal{B}$.
Then the Cayley–Hamilton theorem gives

$$\boxed{\chi_T(T) = 0.}$$

This formulation is independent of coordinates.
Thus every finite-dimensional linear operator satisfies a nonzero polynomial equation.

7.17 Reducing Powers of a Matrix

The Cayley–Hamilton theorem allows high powers of a matrix to be reduced to lower powers.
If

$$\chi_A(X) = X^n + c_{n-1}X^{n-1} + \cdots + c_0,$$

then

$$A^n = -c_{n-1}A^{n-1} - \cdots - c_1 A - c_0 I.$$

Multiplying repeatedly by A, every power A^m with $m \geq n$ can be expressed as a linear combination of

$$I, A, \ldots, A^{n-1}.$$

Example 7.17.1. Suppose

$$A \in M_2(K)$$

has characteristic polynomial

$$\chi_A(X) = X^2 - 5X + 6.$$

Then

$$A^2 - 5A + 6I = 0,$$

so

$$A^2 = 5A - 6I.$$

Hence

$$A^3 = 5A^2 - 6A = 19A - 30I.$$

Thus Cayley–Hamilton converts high matrix powers into finite linear algebra.

7.18 Inverse from the Characteristic Polynomial

Suppose A is invertible. Then the constant term

$$c_0 = (-1)^n \det A$$

of χ_A is nonzero.
From

$$A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I = 0,$$

we obtain

$$A\left(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I\right) = -c_0 I.$$

Therefore

$$\boxed{A^{-1} = -\frac{1}{c_0}\left(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I\right).}$$

Thus the inverse of an invertible matrix can always be expressed as a polynomial in the matrix itself.

7.19 Polynomial Relations and Eigenvalues

Suppose

$$f(A) = 0$$

for some polynomial $f \in K[X]$.
If $v \neq 0$ is an eigenvector of A with eigenvalue λ, then

$$Av = \lambda v.$$

Hence

$$A^k v = \lambda^k v$$

for every $k \geq 0$, and therefore

$$f(A)v = f(\lambda)v.$$

Since $f(A) = 0$,

$$f(\lambda)v = 0.$$

Because $v \neq 0$,

$$f(\lambda) = 0.$$

Thus:

Theorem 7.19.1. If $f(A) = 0$, then every eigenvalue of A is a root of f.

In particular, Cayley–Hamilton implies once again that every eigenvalue is a root of the characteristic polynomial.

7.20 From Eigenvalues to Diagonalization

This chapter has established the chain

$$\boxed{\det(XI - A) \longrightarrow \chi_A(X) \longrightarrow \text{eigenvalues} \longrightarrow \text{eigenspaces.}}$$

The characteristic polynomial records the eigenvalues and their algebraic multiplicities, while the eigenspaces measure how many independent eigenvectors are available.
For each eigenvalue λ,

$$1 \leq m_g(\lambda) \leq m_a(\lambda).$$

If enough linearly independent eigenvectors can be found to form a basis

$$v_1, \ldots, v_n,$$

then relative to this basis the matrix of the operator is

$$\operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

The central question is therefore:

$$\boxed{\text{When does a linear operator possess a basis of eigenvectors?}}$$

This is precisely the problem of diagonalization, which is the subject of the next chapter.

Chapter 8

Diagonalization and Basic Spectral Structure

8.1 Diagonalizable Linear Operators

Let V be an n-dimensional vector space over a field K, and let

$$T \in \mathrm{End}_K(V).$$

The matrix of T depends on the chosen basis. A natural question is whether one can choose a basis in which the matrix becomes especially simple.
The simplest possible form is a diagonal matrix.

Definition 8.1.1. A linear operator $T \in \mathrm{End}_K(V)$ is diagonalizable over K if there exists a basis $\mathcal{B}$ of V such that

$$[T]_{\mathcal{B}}$$

is diagonal.

For a matrix $A \in M_n(K)$, we say that A is diagonalizable over K if there exists an invertible matrix $P \in M_n(K)$ such that

$$P^{-1}AP$$

is diagonal.
Thus

$$\boxed{A \text{ is diagonalizable } \iff A \text{ is similar to a diagonal matrix.}}$$

If A represents T, then A is diagonalizable if and only if T is diagonalizable.

8.2 Eigenbases

Suppose

$$\mathcal{B} = (v_1, \ldots, v_n)$$

is a basis of V consisting of eigenvectors of T, with

$$T(v_i) = \lambda_i v_i.$$

Then

$$[T]_{\mathcal{B}} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

This observation gives the fundamental criterion for diagonalizability.

Theorem 8.2.1. Let $T \in \mathrm{End}_K(V)$, where $\dim V = n$. Then the following are equivalent:

1. T is diagonalizable over K;
2. V has a basis consisting entirely of eigenvectors of T;
3. there exist n linearly independent eigenvectors of T.

Proof. Suppose T is diagonalizable. Then there exists a basis

$$\mathcal{B} = (v_1, \ldots, v_n)$$

such that

$$[T]_{\mathcal{B}} = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

Therefore

$$T(v_i) = \lambda_i v_i$$

for every i, so every basis vector is an eigenvector.
Conversely, if

$$v_1, \ldots, v_n$$

form a basis of eigenvectors with

$$T(v_i) = \lambda_i v_i,$$

then

$$[T]_{\mathcal{B}} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Hence T is diagonalizable. □

Thus diagonalization is equivalent to finding an eigenbasis.

8.3 Diagonalization of a Matrix

Let

$$A \in M_n(K).$$

Suppose

$$v_1, \ldots, v_n$$

are linearly independent eigenvectors of A, satisfying

$$Av_i = \lambda_i v_i.$$

Define

$$P = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}.$$

Since the columns are linearly independent, P is invertible.
Also,

$$AP = \begin{pmatrix} | & & | \\ Av_1 & \cdots & Av_n \\ | & & | \end{pmatrix} = \begin{pmatrix} | & & | \\ \lambda_1 v_1 & \cdots & \lambda_n v_n \\ | & & | \end{pmatrix}.$$

Therefore

$$AP = P \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Multiplying by P^{-1} on the left gives

$$\boxed{P^{-1}AP = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).}$$

Thus the diagonalizing matrix P is formed by placing an eigenbasis of A into its columns.

Remark 8.3.1. The order of the columns of P must agree with the order of the eigenvalues on the diagonal of D.

8.4 A Practical Diagonalization Procedure

To diagonalize

$$A \in M_n(K),$$

one may proceed as follows.

1. Compute the characteristic polynomial

$$\chi_A(X) = \det(XI_n - A).$$

2. Find all eigenvalues

$$\lambda_1, \ldots, \lambda_r$$

in K.

3. For each eigenvalue λ_i, compute the eigenspace

$$E_{\lambda_i}(A) = \ker(A - \lambda_i I_n).$$

4. Find a basis for each eigenspace.

5. Count the total number of independent eigenvectors obtained.

6. If the total number is n, combine these bases into an eigenbasis and form

$$P = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}.$$

Then

$$P^{-1}AP = D$$

is diagonal.

7. If fewer than n independent eigenvectors exist, then A is not diagonalizable.

The central issue is therefore not merely how many eigenvalues exist, but whether the eigenspaces together contain enough independent vectors to form a basis.

8.5 Example of Diagonalization

Consider

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

The characteristic polynomial is

$$\chi_A(X) = \det \begin{pmatrix} X-1 & 0 & 0 \\ 0 & X-2 & 0 \\ -1 & 0 & X-1 \end{pmatrix} = (X-1)^2(X-2).$$

Thus the eigenvalues are 1 and 2.
For $\lambda = 1$,

$$A - I = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Hence

$$E_1(A) = \left\{ \begin{pmatrix} 0 \\ 0 \\ t \end{pmatrix} : t \in K \right\},$$

so

$$\dim E_1(A) = 1.$$

For $\lambda = 2$,

$$A - 2I = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{pmatrix}.$$

Hence

$$E_2(A) = \left\{ \begin{pmatrix} 0 \\ t \\ 0 \end{pmatrix} : t \in K \right\},$$

so

$$\dim E_2(A) = 1.$$

The total number of linearly independent eigenvectors is therefore only

$$1 + 1 = 2 < 3.$$

Hence A is not diagonalizable.
This example shows that the characteristic polynomial may split completely while the matrix still fails to be diagonalizable.

8.6 Distinct Eigenvalues

The theorem from the previous chapter that eigenvectors associated with distinct eigenvalues are linearly independent immediately gives an important sufficient condition.

Theorem 8.6.1. Let $T \in \mathrm{End}_K(V)$, where

$$\dim V = n.$$

If T has n distinct eigenvalues in K, then T is diagonalizable.

Proof. Choose one nonzero eigenvector for each of the n distinct eigenvalues. Eigenvectors associated with distinct eigenvalues are linearly independent, so these n vectors form a basis of V. Therefore T is diagonalizable. □

The converse is false.
For example,

$$I_n$$

is diagonalizable but has only one eigenvalue, namely 1.
Thus

$$\boxed{n \text{ distinct eigenvalues} \Longrightarrow \text{diagonalizable},}$$

but the reverse implication does not hold.

8.7 Eigenspaces for Distinct Eigenvalues

Let

$$\lambda_1, \ldots, \lambda_r$$

be distinct eigenvalues of T, and define

$$E_i = E_{\lambda_i}(T).$$

Theorem 8.7.1. The sum

$$E_1 + \cdots + E_r$$

is direct.

Equivalently,

$$E_1 + \cdots + E_r = E_1 \oplus \cdots \oplus E_r.$$

Proof. Suppose

$$v_1 + \cdots + v_r = 0, \qquad v_i \in E_i.$$

Discard any zero v_i. The remaining nonzero vectors are eigenvectors corresponding to distinct eigenvalues, so they are linearly independent. Therefore every $v_i = 0$.
Hence the sum is direct. □

Consequently,

$$\dim(E_1 \oplus \cdots \oplus E_r) = \sum_{i=1}^{r} \dim E_i.$$

This provides the dimension criterion underlying diagonalization.

8.8 Diagonalizability and Eigenspace Dimensions

Suppose the distinct eigenvalues of T are

$$\lambda_1, \ldots, \lambda_r.$$

Theorem 8.8.1. The following are equivalent:

1. T is diagonalizable;
2. $$V = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_r}(T);$$
3. $$\sum_{i=1}^{r} \dim E_{\lambda_i}(T) = \dim V.$$

Proof. Suppose

$$V = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_r}(T).$$

Choose a basis of each eigenspace. Since the sum is direct and equals V, the union of these bases is a basis of V consisting of eigenvectors. Hence T is diagonalizable.
Conversely, suppose T is diagonalizable. Then V has an eigenbasis. Every vector in that basis belongs to one of the eigenspaces, so the eigenspaces together span V. Since their sum is already known to be direct,

$$V = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_r}(T).$$

The dimension identity follows immediately. □

Thus

$$\boxed{T \text{ diagonalizable } \iff \sum_{\lambda} \dim E_\lambda(T) = \dim V.}$$

8.9 Algebraic and Geometric Multiplicity

Let λ be an eigenvalue of T.
Recall that its algebraic multiplicity is the multiplicity of λ as a root of $\chi_T(X)$, while its geometric multiplicity is

$$m_g(\lambda) = \dim E_\lambda(T).$$

From Chapter 7,

$$1 \leq m_g(\lambda) \leq m_a(\lambda).$$

Suppose the characteristic polynomial splits over K:

$$\chi_T(X) = \prod_{i=1}^{r} (X - \lambda_i)^{m_i}.$$

Then

$$\sum_{i=1}^{r} m_i = n.$$

Theorem 8.9.1. If the characteristic polynomial of T splits over K, then T is diagonalizable if and only if

$$m_g(\lambda_i) = m_a(\lambda_i)$$

for every eigenvalue λ_i.

Proof. Since distinct eigenspaces form a direct sum,

$$\dim \left(E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_r}(T) \right) = \sum_{i=1}^{r} m_g(\lambda_i).$$

If

$$m_g(\lambda_i) = m_a(\lambda_i)$$

for every i, then

$$\sum_{i=1}^{r} m_g(\lambda_i) = \sum_{i=1}^{r} m_a(\lambda_i) = n.$$

Hence the direct sum of the eigenspaces has dimension n, so it equals V. Therefore T is diagonalizable.
Conversely, if T is diagonalizable, then

$$n = \sum_{i=1}^{r} m_g(\lambda_i).$$

Since

$$m_g(\lambda_i) \leq m_a(\lambda_i)$$

for every i, and

$$\sum_{i=1}^{r} m_a(\lambda_i) = n,$$

equality must hold term by term. □

Thus failure of diagonalizability occurs precisely when at least one eigenspace is too small relative to the algebraic multiplicity of its eigenvalue.

8.10 Diagonalizable Matrices and Matrix Powers

Suppose

$$A = PDP^{-1},$$

where

$$D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Then

$$A^2 = PDP^{-1}PDP^{-1} = PD^2P^{-1}.$$

More generally,

$$\boxed{A^m = PD^mP^{-1}.}$$

Since

$$D^m = \operatorname{diag}(\lambda_1^m, \ldots, \lambda_n^m),$$

powers of A become easy to compute.
More generally, for any polynomial f,

$$f(A) = Pf(D)P^{-1},$$

where

$$f(D) = \operatorname{diag}\Big(f(\lambda_1), \ldots, f(\lambda_n)\Big).$$

Thus diagonalization reduces polynomial expressions in a matrix to scalar calculations on its eigenvalues.

8.11 Application to Linear Recurrences

Diagonalization can be used to solve linear recurrence relations.
Consider the Fibonacci-type recurrence

$$a_{n+1} = a_n + a_{n-1}.$$

Define

$$v_n = \begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix}.$$

Then

$$v_n = Av_{n-1}, \qquad A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Therefore

$$v_n = A^n v_0.$$

The eigenvalues of A are the roots of

$$\lambda^2 - \lambda - 1 = 0,$$

namely

$$\lambda_1 = \frac{1+\sqrt{5}}{2}, \qquad \lambda_2 = \frac{1-\sqrt{5}}{2}.$$

Since the eigenvalues are distinct, A is diagonalizable:

$$A = PDP^{-1},$$

where

$$D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

Hence

$$A^n = PD^nP^{-1}.$$

The recurrence has therefore been reduced to powers of the two scalar eigenvalues. This illustrates one of the principal computational advantages of diagonalization.

8.12 Invariant Subspaces

Diagonalization is closely related to the way a linear operator acts on subspaces.

Definition 8.12.1. Let

$$T \in \mathrm{End}_K(V).$$

A subspace $U \subseteq V$ is called T-invariant if

$$T(U) \subseteq U.$$

If U is T-invariant, then the restriction

$$T|_U : U \to U$$

is itself a linear operator.
The trivial subspaces

$$\{0\} \qquad \text{and} \qquad V$$

are always invariant.
Also,

$$\ker T \qquad \text{and} \qquad \mathrm{Im}\, T$$

are T-invariant.
Every eigenspace

$$E_\lambda(T)$$

is invariant.

Proposition 8.12.2. A one-dimensional subspace $U \subseteq V$ is T-invariant if and only if it is generated by an eigenvector of T.

Proof. Suppose

$$U = \mathrm{span}(v), \qquad v \neq 0.$$

If U is invariant, then

$$T(v) \in U,$$

so there exists $\lambda \in K$ such that

$$T(v) = \lambda v.$$

Hence v is an eigenvector.
The converse is immediate. □

Thus eigenvectors are precisely the generators of one-dimensional invariant subspaces.

8.13 Invariant Subspaces and Block Matrices

Suppose $U \subseteq V$ is T-invariant.
Choose a basis

$$u_1, \ldots, u_r$$

of U, and extend it to a basis

$$u_1, \ldots, u_r, v_1, \ldots, v_s$$

of V.
Because

$$T(u_i) \in U,$$

the first r columns of the matrix of T have no components in the directions $v_1, \ldots, v_s$.
Therefore

$$[T] = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}.$$

The block A is the matrix of

$$T|_U : U \to U$$

with respect to the chosen basis of U.
Thus an invariant subspace produces a block upper triangular representation.

8.14 Direct Sums of Invariant Subspaces

Suppose

$$V = V_1 \oplus \cdots \oplus V_r,$$

where each V_i is T-invariant.
Choose a basis $\mathcal{B}_i$ of each V_i, and combine them into a basis

$$\mathcal{B} = \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_r$$

of V.
Then

$$[T]_{\mathcal{B}} = \begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_r \end{pmatrix},$$

where

$$A_i = [T|_{V_i}]_{\mathcal{B}_i}.$$

Thus

$$\boxed{V = V_1 \oplus \cdots \oplus V_r \text{ with each } V_i \text{ invariant}}$$

corresponds to

$$\boxed{[T] \text{ having a block diagonal form.}}$$

This is the natural generalization of ordinary diagonalization.
Ordinary diagonalization corresponds to decomposing V into one-dimensional invariant subspaces.

8.15 Quasi-Diagonalization

Definition 8.15.1. A linear operator $T \in \mathrm{End}_K(V)$ is quasi-diagonalizable if there exists a basis in which its matrix has block diagonal form

$$[T] = \mathrm{diag}(A_1, \ldots, A_r).$$

Equivalently, V can be decomposed as

$$V = V_1 \oplus \cdots \oplus V_r,$$

where every V_i is T-invariant.
The blocks A_i are matrix representations of the restrictions

$$T|_{V_i}.$$

Ordinary diagonalization is the special case in which every

$$\dim V_i = 1.$$

Thus the hierarchy is

$$\boxed{\text{diagonal form} \subseteq \text{block diagonal form} \subseteq \text{general matrix form.}}$$

8.16 Restrictions of Diagonalizable Operators

An important property of diagonalizable operators is inherited by their invariant subspaces.

Theorem 8.16.1. Let $T \in \mathrm{End}_K(V)$, and let $U \subseteq V$ be a T-invariant subspace. If T is diagonalizable, then

$$T|_U$$

is diagonalizable.

Suppose the distinct eigenvalues of T are

$$\lambda_1, \ldots, \lambda_r,$$

with eigenspaces

$$E_i = E_{\lambda_i}(T).$$

Since T is diagonalizable,

$$V = E_1 \oplus \cdots \oplus E_r.$$

For the restriction $T|_U$, its eigenspace corresponding to λ_i is

$$E_{\lambda_i}(T|_U) = E_i \cap U.$$

The invariant subspace U decomposes as

$$U = (E_1 \cap U) \oplus \cdots \oplus (E_r \cap U).$$

Therefore U possesses a basis consisting of eigenvectors of $T|_U$, and the restriction is diagonalizable.

8.17 Diagonalization as a Change of Coordinates

The conceptual meaning of diagonalization can now be summarized.
A matrix A represents a linear operator relative to one basis. Replacing the basis by another basis corresponds to similarity:

$$A \longmapsto P^{-1}AP.$$

Diagonalization asks whether a basis can be chosen so that

$$P^{-1}AP = D$$

is diagonal.
The columns of P are an eigenbasis, and the diagonal entries of D are the corresponding eigenvalues.
Thus

$$\boxed{\text{diagonalization} = \text{finding coordinates adapted to the invariant directions of } T.}$$

A diagonal operator acts independently along each basis direction:

$$T(v_i) = \lambda_i v_i.$$

There is no mixing between different coordinate directions.
This is why diagonal form is so useful both conceptually and computationally.

8.18 Summary of the First Volume

The development of this volume began with basic mathematical language and proceeded through increasingly structural descriptions of linear phenomena:

$$\boxed{\text{Sets and Maps} \longrightarrow \text{Linear Systems} \longrightarrow \text{Matrices} \longrightarrow \text{Vector Spaces}}$$

and then

$$\boxed{\text{Linear Maps} \longrightarrow \text{Determinants} \longrightarrow \text{Eigenvalues} \longrightarrow \text{Diagonalization.}}$$

Linear systems introduced the computational problem. Matrices encoded that problem efficiently. Vector spaces separated linear structure from coordinates, while linear maps explained matrices as coordinate representations of transformations.
Determinants provided the first major invariant under changes of basis. Characteristic polynomials converted questions about matrices into polynomial equations, and eigenvalues identified invariant directions.
Finally, diagonalization showed how a suitable choice of basis can reduce a linear operator to its simplest possible independent components:

$$T(v_i) = \lambda_i v_i.$$

The central principle underlying the entire volume is therefore the distinction

$$\boxed{\text{intrinsic linear object} \qquad \text{versus} \qquad \text{its coordinate representation.}}$$

A vector is represented by coordinates. A linear map is represented by a matrix. Changing coordinates changes the matrix but not the underlying map. Similarity records this

change, while quantities such as rank, determinant, characteristic polynomial, trace, and eigenvalues retain intrinsic information.
Diagonalization succeeds when these invariant directions are rich enough to form an entire basis.
When this fails, a more refined description of the linear operator is required. The continuation of these notes begins precisely from this problem and develops more general canonical forms, most notably the Jordan normal form.

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These notes were reorganized and rewritten for personal review and future reference, but their mathematical framework and overall progression owe much to Professor Hu's textbook.